

Binary Aquila Optimizer for 0–1 knapsack problems

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ABSTRACT

The optimization process entails determining the best values for various system characteristics in order to finish the system design at the lowest possible cost. In general, real-world applications and issues in artificial intelligence and machine learning are discrete, unconstrained, or discrete. Optimization approaches have a high success rate in tackling such situations. As a result, several sophisticated heuristic algorithms based on swarm intelligence have been presented in recent years. Various academics in the literature have worked on such algorithms and have effectively addressed many difficulties. Aquila Optimizer (AO) is one such algorithm. Aquila Optimizer (AO) is a recently suggested heuristic algorithm. It is a novel population-based optimization strategy. It was made by mimicking the natural behavior of the Aquila. It was created by imitating the behavior of the Aquila in nature in the process of catching its prey. The AO algorithm is an algorithm developed to solve continuous optimization problems in their original form. In this study, the AO structure has been updated again to solve binary optimization problems. Problems encountered in the real world do not always have continuous values. It exists in problems with discrete values. Therefore, algorithms that solve continuous problems need to be restructured to solve discrete optimization problems as well. Binary optimization problems constitute a subgroup of discrete optimization problems. In this study, a new algorithm is proposed for binary optimization problems (BAO). The most successful BAO-T algorithm was created by testing the success of BAO in eight different transfer functions. Transfer functions play an active role in converting the continuous search space to the binary search space. BAO has also been developed by adding candidate solution step crossover and mutation methods (BAO-CM). The success of the proposed BAO-T and BAO-CM algorithms has been tested on the knapsack problem, which is widely selected in binary optimization problems in the literature. Knapsack problem examples are divided into three different benchmark groups in this study. A total of sixty-three low, medium, and large scale knapsack problems were determined as test datasets. The performances of BAO-T and BAO-CM algorithms were examined in detail and the results were clearly shown with graphics. In addition, the results of BAO-T and BAO-CM algorithms have been compared with the new heuristic algorithms proposed in the literature in recent years, and their success has been proven. According to the results, BAO-CM performed better than BAO-T and can be suggested as an alternative algorithm for solving binary optimization problems.

1. Introduction

Real-world applications and issues in artificial intelligence and machine learning are typically discrete, unconstrained, or discrete in nature (Hajipour et al., 2014). Using typical mathematical-based programming approaches, it is challenging to identify optimal solutions. Optimization problems can be found in all fields of science. Various optimization techniques have been introduced in recent years to enhance the performance of various systems and minimize computing expenses. They have been applied to solve various problems. Metaheuristic algorithms have garnered a lot of interest in recent years and have been utilized to handle a variety of optimization issues. In general, we can divide Metaheuristic algorithms into four categories, Swarm Intelligence algorithms (Dorigo et al., 1996) (SI), Evolutionary Algorithms (Holland, 1992) (EAs), Physics-based algorithms (Rashedi

et al., 2009) (PhAs), and Human-based algorithms (Kumar et al., 2018). In general, all metaheuristic algorithms have common characteristics, such as the search process, which is divided into two stages: diversification (exploration) and intensification (exploitation). In this study, crossover and mutation methods were chosen to ensure diversity and exploitation. Crossover and mutation operators have been used in candidate solution generation in many different studies (Lim et al., 2017; Patil and Pawar, 2015).

Binary optimization problems constitute a different subgroup of discrete optimization problems. Binary optimization search space is expressed with $\{0, 1\}$ while discrete optimization search space is expressed with real discrete values. Expressing the search space with binary values offers convenience in solving many real-world problems. Transfer functions are frequently used in the literature to convert continuous search space to binary search space. Various transfer functions

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are available in the literature. The most preferred of these are S and V-shaped transfer functions. The transfer functions selected in this study are shown in the second section. In this study, the 0–1 knapsack optimization problem is chosen as a binary optimization problem. There are many binary optimization problems in the literature. Some of them are uncapacitated facility layout problems, Feature Selection (FS) problems, Knapsack Problems (KP), etc. (Baş and Ülker, 2020a,b). Knapsack issues (KP) derive their importance from appearance in many real-world applications. These problems are commonly seen in project portfolio selection, energy minimization, cargo loading problems, etc. (Cho, 2019; Müller et al., 2015; Bas, 2012). Therefore, tackling KP allows many applications to thrive and grow. But KP is an NP-hard problem, so it is difficult to find a polynomial-time solution. KP is a combinatorial optimization problem in which we try to find the optimal solution among a finite set of solutions. As the size of the KP increases, the time it takes for exact methods to search for the optimal solution increases exponentially. Precise methods such as exhaustive search or branch-and-bounds consume enormous computational resources in terms of space and time, especially for large-scale KPs. Therefore, the need to find a near-optimal solution in an acceptable time is a trend that many researchers follow to solve KP (Abdel-Basset et al., 2021). Many studies in the literature solve the knapsack problem. Hakli proposed the BinEHO algorithm for the 0–1 knapsack problem. It is a new binary variant based on an elephant herding optimization algorithm (Hakli, 2020). Abdel-Basset et al. proposed a Binary Equilibrium Optimization Algorithm for the 0–1 knapsack problems (Abdel-Basset et al., 2021). They demonstrated the success of BEO by updating BEO according to eight different transfer functions. Kulkarni and Shabir proposed solving the 0–1 knapsack problem using Cohort Intelligence Algorithm (Kulkarni and Shabir, 2016). Wu et al. combined the symbiotic search algorithm with the harmony search for solving small and large-scale knapsack problems (Wu et al., 2018). Huang et al. Proposed a Binary Multi-Scale Quantum Harmonic Oscillator Algorithm for the 0–1 knapsack problem with the genetic operator (Huang et al., 2019). Feng et al. proposed a novel binary monarch butterfly optimization (BMBO) method, intended for addressing the 0–1 knapsack problem (0–1 KP) (Feng et al., 2017). They used two tuples of real-valued vectors and binary vectors to represent monarch butterfly individuals in BMBO. Monarch butterfly optimization works directly on real-valued vectors, and solutions are represented by binary vectors (Feng et al., 2017). Feng et al. proposed generalized opposition-based learning (OBL) monarch butterfly optimization with Gaussian perturbation (OMBO) for the 0–1 knapsack problem (Feng et al., 2018a). In the late stage of evolution, the OBL approach is employed on half of the population, and Gaussian perturbation is applied to a portion of the individuals with poor fitness in each evolution. OBL ensures that OMBO's convergence speed is faster, and Gaussian perturbation prevents it from being trapped at a local optimum (Feng et al., 2018a). Feng et al. offered a multi-strategy monarch butterfly optimization algorithm for the discount 0–1 knapsack problem (Feng et al., 2018b). Feng et al. presented a monarch butterfly optimization algorithm with differential evolution for the discount 0–1 knapsack problem (Feng et al., 2018c). Feng et al. proposed a novel global firefly algorithm (GFA) is proposed for solving randomized time-varying knapsack problems (RTVKP) (Feng et al., 2018d).

In this study, a new natural inspiration-based meta-heuristic optimization algorithm called Aquila Optimizer (AO) is introduced. The AO algorithm was first proposed by Abualigah et al. in 2021. The proposed AO algorithm simulates the behavior of Aquila during the hunt, showing the actions of each step of the hunt. The AO algorithm has not yet been proposed for different optimization problems, except for continuous optimization problems, in the literature. In this paper, the success of the AO algorithm on binary optimization problems, which is one of the discrete optimization problem types, is tested. In this study, continuous search space is converted to binary search space by using eight different transfer functions AO and Binary AO (BAO)

is proposed. A popular binary optimization problem, the knapsack problem, was chosen to test the success of BAO. Knapsack datasets of 63 different sizes are grouped into 3 different groups. In the first stage, BAO was created and tested in three different knapsack benchmark samples according to eight different transfer functions (BAO1, BAO2, BAO3, BAO4, BAO5, BAO6, BAO7, and BAO8). The most successful BAO variation was named BAO-T. In the second stage, a new candidate generation stage was developed by adding crossover and mutation operators to BAO. Thus, BAO's exploration and exploitation capabilities have been enhanced. The proposed BAO-T and BAO-CM were compared on the first and second knapsack benchmark samples. The results are interpreted and shown with graphics. In the third stage, the proposed BAO-T and BAO-CM algorithms were compared with the new heuristic algorithms that solve the 0–1 knapsack problem in the literature for the first and second knapsack samples in recent years. The success of BAO is shown by comparing the results according to different comparison criteria (best, worst, mean, standard deviation, and gap). In the fourth stage, the success of BAO-CM for the third knapsack benchmark samples was demonstrated and compared with various new heuristic algorithms in the literature. Thus, BAO-CM has proven successful not only in low- and medium-sized knapsack datasets but also in large-scale knapsack datasets.

1.1. The contributions of the study to the literature

This study presented the following contributions to the literature.

- For the first time, the AO algorithm was updated to solve binary optimization problems by updating its original version, and binary AO (BAO) was proposed to the literature.
- Parameter settings affecting the performance of BAO have been tested in detail and the most appropriate values have been selected.
- Eight different BAO variations were created according to the eight different S and V-shaped transfer functions of BAO. These variations were tested on the first knapsack samples and the most successful transfer function was determined. The most successful BAO variation was named BAO-T.
- To improve the exploitation and exploration capabilities of BAO, the new candidate generation stage of BAO was developed by adding crossover and mutation operators and BAO-CM was proposed.
- Variations of BAO-T and BAO-CM were compared in detail with various heuristic algorithms selected from the literature according to the best, worst, mean, standard deviation, and gap criteria.
- The success of BAO-CM has been proven by running BAO-CM not only on low and medium-scaled knapsack samples but also on large-scaled knapsack samples.

This work follows as follows: Section 2 shows the structure of AO, Binary AO (BAO), Binary AO with crossover and mutation (BAO-CM), transfer functions, and the knapsack problem. In Section 3, the performances of Binary AO with transfer function (BAO-T) and the performances of Binary AO with crossover and mutation (BAO-CM) are shown for low, medium, and large-scaled knapsack problems. BAO-T and BAO-CM have been compared with various new heuristic algorithms in the literature. Comparisons with the latest methods are presented. Finally, Section 4 concludes that the study was successful in various aspects.

2. Material and method

2.1. Aquila Optimizer (AO)

The Aquila is one of the most popular birds of prey in the Northern Hemisphere. The Aquila is dark brown in color with light golden brown feathers on the back of its neck (Abualigah et al., 2021). The Aquila

captures a variety of prey, mostly rabbits, hares, marmots, squirrels, and other land animals, using its speed and agility, as well as its robust feet and huge, sharp claws. Aquila has areas that can be up to 200 km². They construct massive nests on mountains and other high areas. Reproductive activities occur in the spring; they are monogamous and can live together for many years, if not their entire existence. Females lay up to four eggs, which are subsequently incubated for six weeks. In most cases, one or two infants survive to feather in around 12 weeks. These young Aquila normally get complete confidence in the fall and then migrate on to large-scale foraging for themselves.

Aquila is one of the most researched birds in the world because of its hunting bravery. When male Aquila hunted alone, he caught more prey. Aquila hunts squirrels, rabbits, and a variety of other creatures using their speed and keen talons. They have even been identified as a predator of full-grown deer. The ground squirrel is the most significant animal in Aquila's diet.

There are four accepted hunting methods. The following items describe Aquila's hunting techniques.

- The first method, high soar with a vertical stoop, is used to hunt birds in flight, with the Aquila rising high above the ground.
- The second approach, contour combat with brief glide assault, is considered Aquila's most commonly utilized method, in which the Aquila climbs at a low height over the ground. Whether the prey is running or flying, it is then pursued carefully.
- The third option is to fly low and attack with a leisurely descent. In this case, the Aquila dives to the ground before launching an attack on the prey.
- The fourth way is walking and grasping prey, in which the Aquila moves across the ground and attempts to draw its prey with it.

As a result, the Aquila is a very clever and skilled hunter. It can provide light on people's troubles. The above-mentioned hunting strategies served as the primary motivation for the AO algorithm.

2.1.1. Mathematical model of AO

The AO algorithm mimics the Aquila's behavior throughout the hunt, displaying the activities at each stage of the hunt. As a result, the AO algorithm's optimization phases are represented by four approaches. They are as follows:

- (1-) Choosing a search space by high soar with a vertical stoop,
- (2-) Exploring within a diverge search space by contour flight with a brief glide attack,
- (3-) Exploring within a converged search space by low flight with a gradual descent attack,
- (4-) Swooping by strolling and catching prey.

The AO algorithm may switch between exploration and exploitation stages.

If $t \leq \left(\frac{2}{3}\right) * T$ the exploration steps will be exciting; otherwise, the exploitation steps will be carried out. The following is a mathematical model of the AO.

Expanded exploration method (X_1):

The Aquila detects the prey area and picks the ideal hunting location by high flying with the vertical stoop in the first way (X_1). Here, the AO extensively explores the search region from a high altitude to discover where the prey is. Fig. 1 depicts the vertical stoop behavior of the Aquila high soar. Eqs. (1)–(2) represent this behavior formally.

$$X_1(t+1) = X_{best}(t) \times \left(1 - \frac{t}{T}\right) + (X_M(t) - X_{best}(t) * rand) \quad (1)$$

$$X_M(t) = \frac{1}{N} \sum_{i=1}^N X_i(t), \quad \forall j = 1, 2, \dots, Dim \quad (2)$$

where $X_1(t+1)$ is the solution of the next iteration of t , which is generated by the first search method (X_1). $X_{best}(t)$ is the best-obtained solution until the t th iteration, this reflects the approximate place of

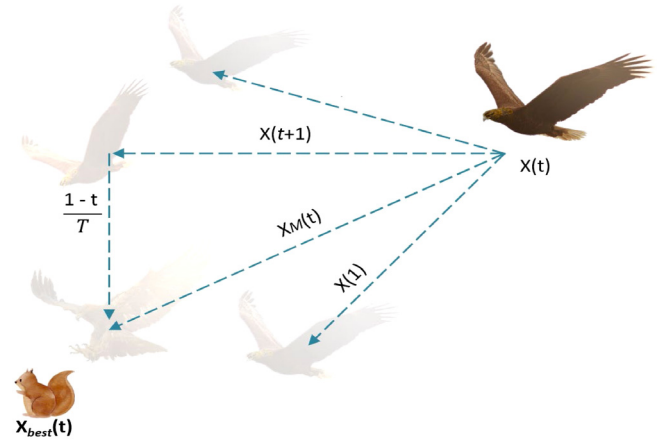


Fig. 1. The behavior of the Aquila high soar with the vertical stoop (Abualigah et al., 2021).

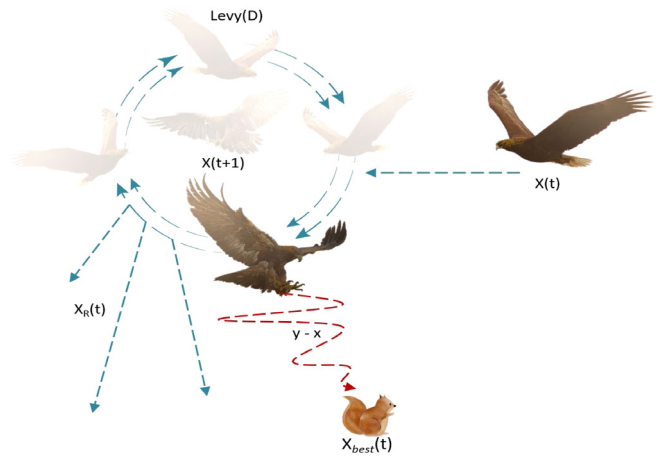


Fig. 2. The behavior of the Aquila contour flight with a short glide attack (Abualigah et al., 2021).

the prey. This equation $\left(1 - \frac{t}{T}\right)$ is used to control the expanded search (exploration) through the number of iterations. $X_M(t)$ denotes the location's mean value of the current solutions connected at t th the iteration. $rand$ is a random value between 0 and 1. t and T present the current iteration and the maximum number of iterations, respectively. Dim is the dimension size of the problem and N is the number of candidate solutions (population size).

Narrowed exploration method (X_2):

In the second method (X_2), when the prey area is found from a high flight, Aquila circles over the target prey, prepare the ground, and then attacks. This method is called contour flying with a short glide attack. Here, the AO narrowly searches the selected area of the target prey in preparation for the attack. Fig. 2 shows the behavior of Aquila contour flight with a short glide attack. This behavior is mathematically presented as in Eqs. (3)–(10).

$$X_2(t+1) = X_{best}(t) \times Levy(D) + X_R(t) + (y - x) * rand \quad (3)$$

where $X_2(t+1)$ is the solution of the next iteration of t . D is the dimension area and $Levy(D)$ is the levy flight distribution function, $X_R(t)$ is a random solution taken in the range of $[1 \ N]$ at the i th iteration.

$$Levy(D) = 0.01 \times \frac{u \times \sigma}{|v|^{\frac{1}{\beta}}} \quad (4)$$

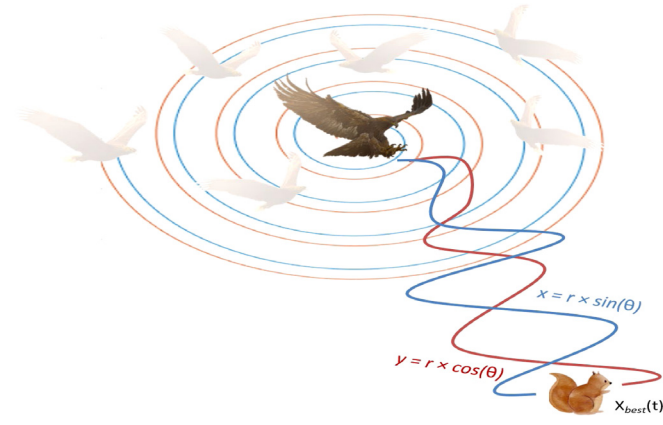


Fig. 3. The behavior of the Aquila low flight with a slow descent attack (Abualigah et al., 2021).

where u and v are random numbers between 0 and 1.

$$\sigma = \frac{\Gamma(1 + \beta) \times \sin\left(\frac{\pi\beta}{2}\right)}{\Gamma\left(\frac{1+\beta}{2}\right) \times \beta \times 2^{\left(\frac{\beta-1}{2}\right)}} \quad (5)$$

where β is a constant value fixed to 1.5.

$$y = r \times \cos(\theta) \quad (6)$$

$$x = r \times \sin(\theta) \quad (7)$$

$$r = r_1 + U \times D_1 \quad (8)$$

$$\theta = -\omega \times D_1 + \theta_1 \quad (9)$$

$$\theta_1 = \frac{3 \times \pi}{2} \quad (10)$$

r_1 takes a value between 1 and 20 for fixed the number of search cycles, and U is a small value fixed to 0.00565. D_1 is an integer number from 1 to the length of the search space (Dim), and ω is a small value fixed to 0.005.

Expanded exploitation method (X_3):

In the third method (X_3), when the prey area is correctly identified and the Aquila is ready to land and attack, the Aquila descends vertically with a frontal attack to explore the prey's response. This method is called low flight with slow landing attack. Here AO uses the selected area of the target to approach prey and attack. Fig. 3 shows Aquila's low-flying behavior with a slow attack. This behavior is mathematically presented as in Eq. (11).

$$X_3(t+1) = (X_{best}(t) - X_M(t)) \times \alpha - rand + ((UB - UL) \times rand + LB) \times \delta \quad (11)$$

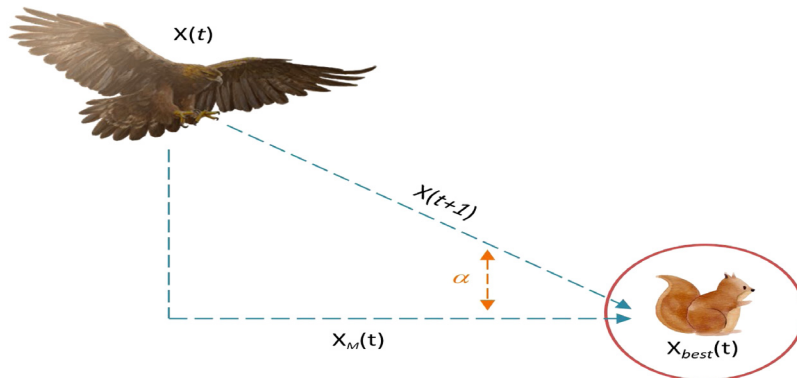


Fig. 4. The behavior of the Aquila low flight with slow descent attack (Abualigah et al., 2021).

where $X_3(t+1)$ is the solution of the next iteration of t . $X_{best}(t)$ refers to the approximate location of the prey until i th iteration (the best-obtained solution), and $X_M(t)$ denotes the mean value of the current solution at t th iteration. $rand$ is a random value between 0 and 1. α and δ are the exploitation adjustment parameters fixed to a small value (0.1). LB denotes the lower bound and UB denotes the upper bound of the given problem.

Narrowed exploitation method (X_4):

In the fourth method (X_4), when Aquila approaches its prey, Aquila attacks its prey over land according to its stochastic movements. This method is called walk-and-catch hunting. Here AO attacks its prey in the last position. Figs. 4, 5, and 6 show Aquila's low flight with slow landing attack, Aquila's walking and prey catching behavior, and the effects of quality function (QF), G_1 , and G_2 on AO behavior. These behaviors are presented mathematically as in Eqs. (12)–(15).

$$X_4(t+1) = QF \times X_{best}(t)$$

$$- (G_1 \times X(t) \times rand) - G_2 \times Levy(D) + rand \times G_1 \quad (12)$$

$$QF(t) = t^{\frac{2 \times rand() - 1}{(1-t)^2}} \quad (13)$$

$$G_1 = 2 \times rand() - 1 \quad (14)$$

$$G_2 = 2 \times \left(1 - \frac{t}{T}\right) \quad (15)$$

where $X_4(t+1)$ is the solution of the next iteration of t . QF denotes a quality function used to equilibrium the search strategies (Eq. (13)). G_1 denotes various motions of the AO that are used to track the prey during the elope (eqrefTeXFolio:eq14). G_2 presents decreasing values from 2 to 0, which denote the flight slope of the AO that is used to follow the prey during the elope from the first location (I) to the last location (t) (Eq. (15)). $X(t)$ is the current solution at the t th iteration. $rand$ is a random value between 0 and 1. t and T present the current iteration and the maximum number of iterations.

To summarize, a random set of candidate solutions, called population, is generated in AO. To strike a balance between AO's exploration strategies (ie exploration and exploitation), four different search strategies are provided for exploration and exploitation (i.e. extended exploration, narrowed exploration, extended exploitation, and narrowed exploitation). Finally, the AO's search process is terminated when the last criterion is met. The pseudocode of AO is detailed in Algorithm 1. Fig. 7 shows the flowchart of the AO.

2.2. Binary Aquila Optimizer (BAO)

In this subsection, the Aquila Optimizer (AO) algorithm is updated again to solve the binary optimization problems (Abualigah et al., 2021). The standard AO is designed to solve the continuous optimization problem, and the knapsack problem is considered a separate problem, so it must be redesigned to solve this problem. One of the transfer functions shown in Table 1 is used to convert the standard

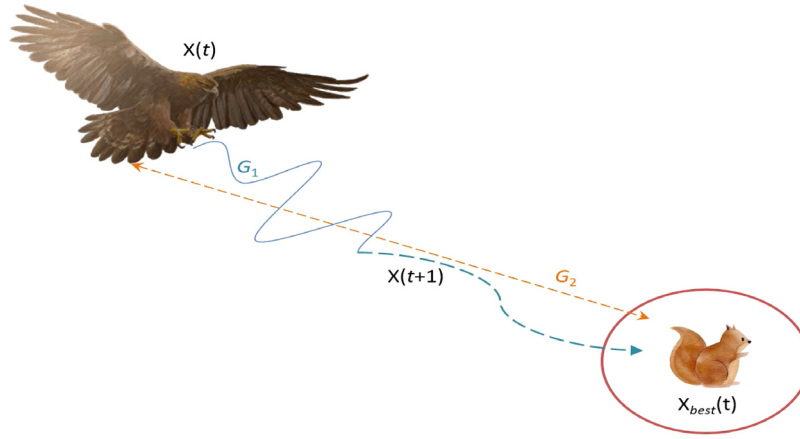


Fig. 5. The behavior of the Aquila walks and grabs prey (Abualigah et al., 2021).

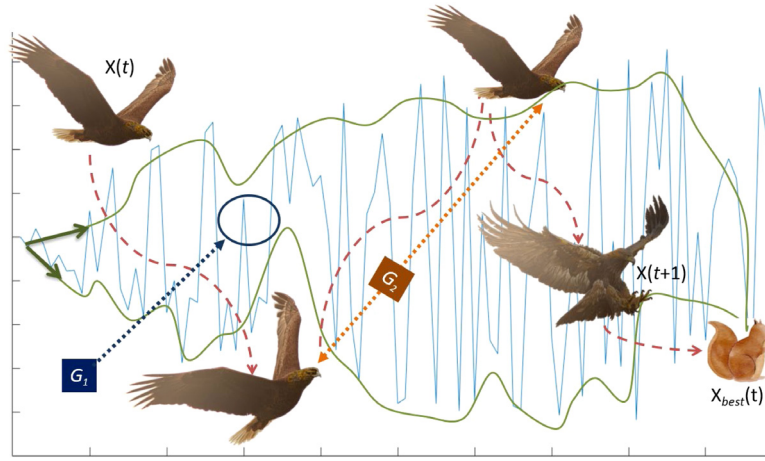


Fig. 6. The effects of the quality function (QF), G_1 , and G_2 on the behavior of the AO (Abualigah et al., 2021).

AO to a binary version, where the continuous value is returned from the standard algorithm and converted to “0” or “1”. A value of “0” indicates the item retrieved from the knapsack and this item is removed from the knapsack. A value of “1” indicates the item selected in the knapsack and this item is placed in the knapsack.

2.2.1. Initialization

The search space consists of N (population size) particles. Each particle has a Dim (dimension) size. Every dimension of every particle consists of real values. These values are converted to binary $\{0, 1\}$ structures with Eq. (16). The first shape of the search space used in the 0–1 knapsack solution is shown in Fig. 8 as a representation.

$$X_{i,j} = \begin{cases} 1 & \text{if } X_{i,j} > 0.5 \\ 0, & \text{otherwise} \end{cases} \quad i = 1, 2, 3, \dots, N; \quad j = 1, 2, 3, \dots, Dim \quad (16)$$

2.2.2. Fitness function

The fitness function is a function that defines the problem in all meta-heuristic algorithms. The knapsack problem is a maximization problem. The core of the search space is searched with the fitness function and a maximum value is tried to be obtained. Used with AO to solve the 0–1 knapsack problem, the fitness function is calculated as the sum of the profits of the items contained in the knapsack. The knapsack problem is shown in Eqs. (16) and (17). Where w_i and x_i are the weight and the status, 0, or 1, of each item in the knapsack, and W is the knapsack capacity. The aim here is to achieve maximum profit

Table 1

S-shaped and V-shaped transfer functions (Abualigah et al., 2021).

S-shaped		V-shaped	
Name	Transfer functions	Name	Transfer functions
S1	$F(a) = \frac{1}{1+e^{-2a}}$	V1	$F(a) = \left \operatorname{erf} \left(\frac{\sqrt{\pi}}{2} a \right) \right $
S2	$F(a) = \frac{1}{1+e^{-a}}$	V2	$F(a) = \tanh(a) $
S3	$F(a) = \frac{1}{1+e^{-a/2}}$	V3	$F(xa) = \left \frac{(a)}{\sqrt{1+a^2}} \right $
S4	$F(a) = \frac{1}{1+e^{-a/3}}$	V4	$F(a) = \left \frac{2}{\pi} \arctan \left(\frac{\pi}{2} a \right) \right $

in the products that will be placed in the backpack, provided that they do not exceed the W capacity.

2.2.3. New candidate production methods

To solve binary optimization problems with the AO algorithm, the search strategy must be adapted to the binary search space because it produces continuous values. As mentioned in the introduction, there are many ways to adapt to the binary search space, such as rounding, transfer functions, crossover, and mutation. The most important point of this conversion process is to preserve and not weaken the search capability of the optimization algorithm. In this study, firstly, the most suitable transfer function selection process for the basic binary AO (BAO) is done. There are eight different transfer functions divided into two classes that are frequently used in the literature, S-Shaped, and V-Shaped (Mirjalili and Lewis, 2013). These functions allow continuous

Algorithm 1: Aquila Optimizer Algorithm (AO) (Abualigah et al., 2021)

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1: Initialization the population  $X$  ( $i=1, 2, 3, \dots, N$ ) ( $X$ = the current solution ) ( $N$ =population size)
2: Initialize the parameters (i.e.,  $\alpha$ ,  $\delta$ ,  $T$ , etc).
3: while ( $t \leq T$ ) do
4:   Calculate the fitness function values and determine the best-obtained solution ( $X_{best}(t)$ )
5:   for ( $i=1, 2, \dots, N$ ) do
6:     Update the mean value of the current solution  $X_M(t)$  using Equation (2).
7:     Update the  $x$ ,  $y$ ,  $G_1$ ,  $G_2$ , Levy( $D$ ), etc.
8:     if  $t \leq \left(\frac{2}{3}\right) * T$  then
9:       if  $rand \leq 0.5$  then
10:        { Expanded exploration ( $X_1$ ) }
11:        Update the current solution using Equation (1).
12:        if Fitness( $X_i(t+1)$ ) < Fitness( $X(t)$ ) then
13:           $X(t)=X_i(t+1)$ 
14:          if Fitness( $X_i(t+1)$ ) < Fitness( $X_{best}(t)$ ) then
15:            The  $X_{best}(t)=X_i(t+1)$ 
16:          end if
17:        end if
18:      else
19:        { Narrowed exploration ( $X_2$ ) }
20:        Update the current solution using Equation (3).
21:        if Fitness( $X_2(t+1)$ ) < Fitness( $X(t)$ ) then
22:           $X(t)=X_2(t+1)$ 
23:          if Fitness( $X_2(t+1)$ ) < Fitness( $X_{best}(t)$ ) then
24:            The  $X_{best}(t)=X_2(t+1)$ 
25:          end if
26:        end if
27:      end if
28:    else
29:      if  $rand \leq 0.5$  then
30:        { Expanded exploitation ( $X_3$ ) }
31:        Update the current solution using Equation (11).
32:        if Fitness( $X_3(t+1)$ ) < Fitness( $X(t)$ ) then
33:           $X(t)=X_3(t+1)$ 
34:          if Fitness( $X_3(t+1)$ ) < Fitness( $X_{best}(t)$ ) then
35:            The  $X_{best}(t)=X_3(t+1)$ 
36:          end if
37:        end if
38:      else
39:        { Narrowed exploitation ( $X_4$ ) }
40:        Update the current solution using Equation (12).
41:        if Fitness( $X_4(t+1)$ ) < Fitness( $X(t)$ ) then
42:           $X(t)=X_4(t+1)$ 
43:          if Fitness( $X_4(t+1)$ ) < Fitness( $X_{best}(t)$ ) then
44:            The  $X_{best}(t)=X_4(t+1)$ 
45:          end if
46:        end if
47:      end if
48:    end if
49:  end for
50: end while
51: The optimum solution ( $X_{best}$ ).

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values to be converted to “0” or “1” values. These transfer functions take a real value as input and convert it to a value between 0 and 1 to produce output. Each transfer function has a special formula to perform this transformation. These formulas are shown in Table 1. In this study, the success of AO in eight different transfer functions (BAO1, BAO2, BAO3, BAO4, BAO5, BAO6, BAO7, and BAO8) was tested and demonstrated on twenty different small-scale knapsack datasets. The most successful transfer function has been determined.

After selecting the most successful transfer function for the development of local and global search capabilities of the binary AO algorithm, new candidate solutions were produced by applying crossover and mutation operations on the obtained binary solutions. These new candidate solutions are compared with the existing solutions and it is predicted to replace them with the current population members. Crossover is performed with the best individual of the population, thus improving local search ability. Eq. (17) shows the crossover operation. The important thing in this crossover operation is how many dimensions will be updated with this equation. In the

basic AO algorithm, if all dimensions are updated simultaneously, each individual in the herd is similar to the best individual. On the other hand, updating a dimension leads to slow convergence. The number of dimensions to be crossed is calculated as $S_{Dim} \times Dim$ (size amount selected dimension = S_{Dim} and Dim =dimension). Since the size of each knapsack problem is different, it was considered appropriate to calculate it in this way. The dimensions to be crossover are chosen randomly and are not equal to each other. Assuming the Dimension (Dim) and S_{Dim} numbers are 10 and 0.5 respectively, Fig. 9 shows the crossover process for each population individual.

$$X_{i,new}^{j,dim} = X_{best}^{j,dim} \quad (17)$$

The mutation process was carried out to ensure diversity in the population. It was carried out by changing a randomly selected size value in each population individual. The mutation process is shown in Fig. 10.

The population update operator is adapted to binary values with these two operations, keeping the principles and advantages of the

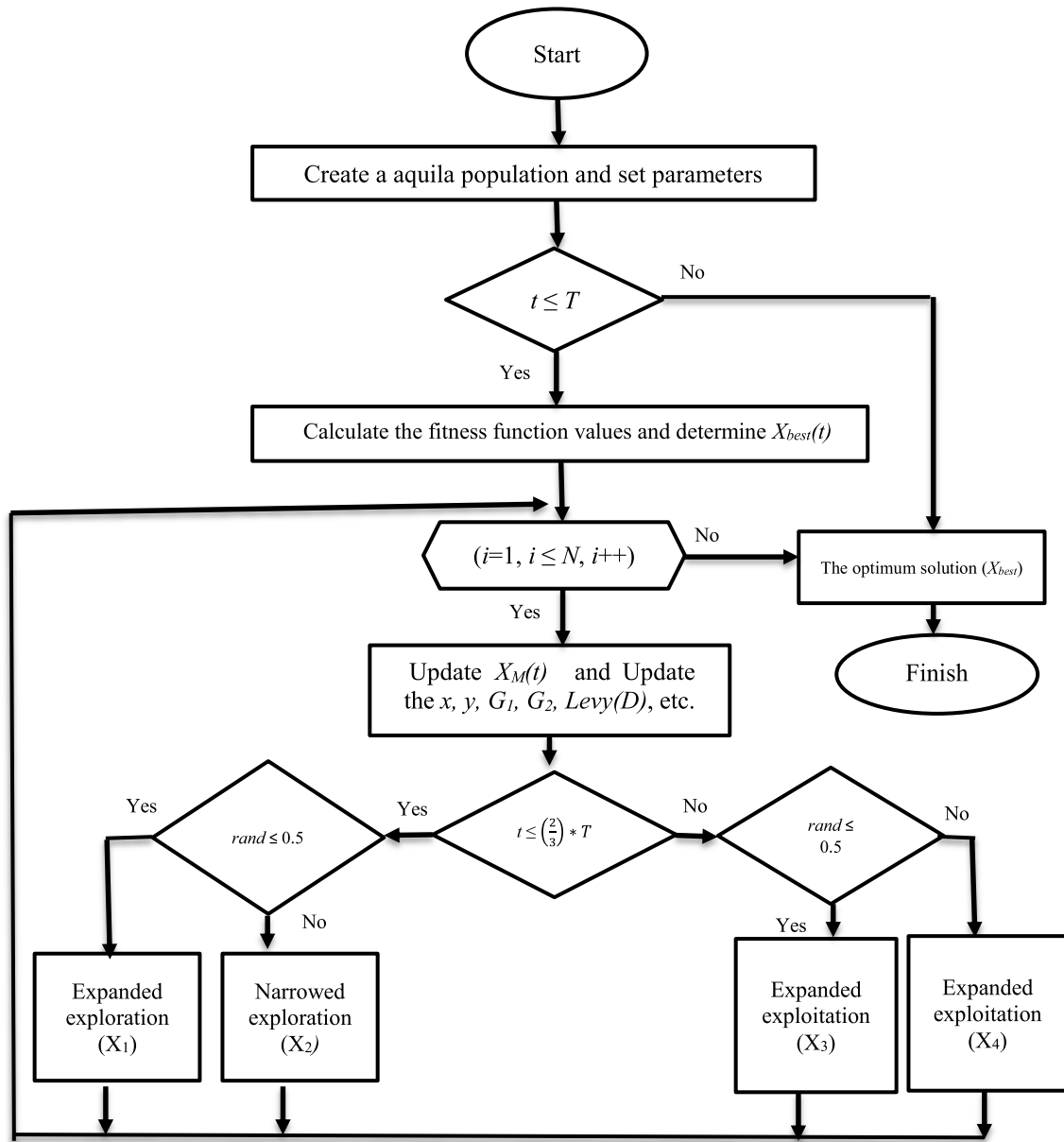


Fig. 7. The flowchart of the AO.

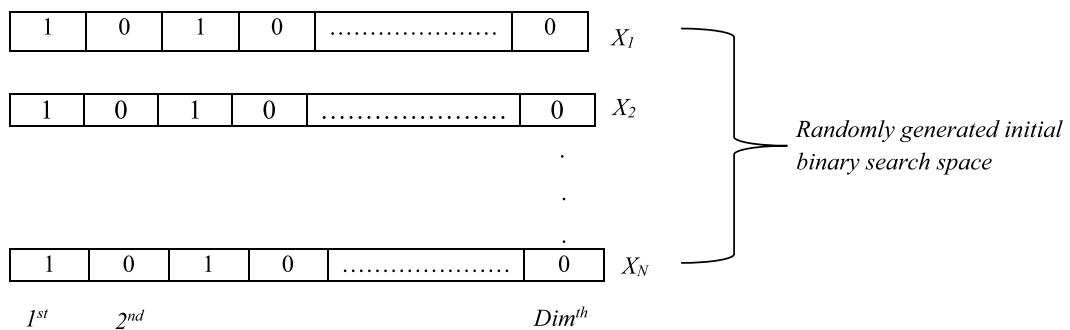


Fig. 8. The initial search space of the binary AO.

Assume that $Dim=10$ and $S_{Dim}=0.5$

1	2	3	4	5	6	7	8	9	10	
0	1	1	0	1	0	0	1	0	1	Aquila

Randomly selected dimensions= {1, 3, 6, 9, 10}

1	2	3	4	5	6	7	8	9	10	
1	0	1	0	1	1	1	1	0	0	Aquila

Updating the aquila with Eq. (17).

1	2	3	4	5	6	7	8	9	10	
1	1	1	0	1	1	0	1	0	0	Aquila

Fig. 9. The crossover process for BAO.

1	2	3	4	5	6	7	8	9	10	
1	0	1	0	1	1	1	1	0	0	Aquila

Randomly selected dimension= 8

1	2	3	4	5	6	7	8	9	10	
1	1	1	0	1	1	0	0	0	0	Aquila

Fig. 10. The mutation process for BAO.

basic AO algorithm. As AO adapts to the binary search space, it is much more simplified.

2.2.4. Evaluation

Fitness function is applied for new candidate individuals produced after crossover and mutation processes and fitness solutions are obtained. By comparing the fitness functions obtained from the old individuals of the population with these new fitness values, candidate solutions with better fitness values replace the old individuals. Thus, the population updates itself in each iteration. The binary version of AO with transfer function (BAO-T) is introduced in Algorithm 2. The binary version of AO with crossover and mutation (BAO-CM) is introduced in Algorithm 3.

2.3. 0–1 Knapsack problem

A typical NP-hard problem, the 0–1 knapsack problem, is defined as follows: Suppose we have N objects. Let the W value indicate the capacity of the knapsack. Each object i has a weight (w_i) and a profit value (p_i). The main goal is to maximize the overall profit of the knapsack without exceeding the limit of the knapsack's capacity. The mathematical model of the problem is very simple and is formulated as follows:

$$\text{Maximize } f(x) = \sum_{i=1}^N x_i p_i X_{i,new}^{j,dim} = X_{best}^{j,dim} \quad (18)$$

$$\text{subject to } \sum_{i=1}^N x_i w_i \leq W, x_i \in \{0, 1\}, i = 1, 2, 3, \dots, N \quad (19)$$

where x_i represents binary decision variables. x_i stores information about the objects that should be in the knapsack and the objects that should not be.

3. Experimental results

In this paper, the success of the AO algorithm on binary optimization problems, which is one of the discrete optimization problem types, is tested. In this study, continuous search space is converted to binary search space by using eight different transfer functions AO and Binary AO (BAO) is proposed. A popular binary optimization problem, the knapsack problem, was chosen to test the success of BAO. Knapsack datasets of sixty-three different sizes are grouped into three different groups. In the first stage, BAO was created and tested in three different knapsack benchmark samples according to eight different transfer functions (BAO1, BAO2, BAO3, BAO4, BAO5, BAO6, BAO7, and BAO8). The most successful BAO variation was named BAO-T. In the second stage, a new candidate generation stage was developed by adding crossover and mutation operators to BAO. Thus, BAO's exploration and exploitation capabilities have been enhanced. The proposed BAO-T and BAO-CM were compared on the first and second knapsack benchmark samples. The results are interpreted and shown with graphics. In the third stage, the proposed BAO-T and BAO-CM algorithms were compared with the new heuristic algorithms that solve the 0–1 knapsack problem in the literature for the first and second knapsack samples in recent years. The success of BAO is shown by comparing the results according to different comparison criteria (best, worst, mean, standard deviation, and gap). In the fourth stage, the success of BAO-CM for the third knapsack benchmark samples was demonstrated and compared with various new heuristic algorithms in the literature. Thus, BAO-CM has

Algorithm 2: Binary Aquila Optimizer Algorithm (BAO-T) (with transfer function)

```

1: Initialization the population  $X_{i,N}$  ( $i=1, 2, 3, \dots, N$ ) ( $X$ = the current solution ) ( $N$ =population size)
2: Initialize the parameters (i.e.,  $\alpha$ ,  $\delta$ ,  $T$ , etc).
3: while ( $t \leq T$ ) do
4:   Update the population  $X$  values using Equation (16).
5:   Calculate the fitness function values and determine the best-obtained solution ( $X_{best}(t)$ )
6:   for ( $i=1, 2, \dots, N$ ) do
7:     Update the mean value of the current solution  $X_M(t)$  using Equation (2).
8:     Update the  $x$ ,  $y$ ,  $G_i$ ,  $G_2$ ,  $Levy(D)$ , etc.
9:     if  $t \leq (\frac{2}{3}) * T$  then
10:      if  $rand \leq 0.5$  then
11:        { Expanded exploration ( $X_1$ ) }
12:        Update the current solution using Equation (1).
13:        Apply one of the eight transfer functions on population  $X_1$ 
14:        if  $Fitness(X_1(t+1)) < Fitness(X(t))$  then
15:           $X(t) = X_1(t+1)$ 
16:          if  $Fitness(X_1(t+1)) < Fitness(X_{best}(t))$  then
17:            The  $X_{best}(t) = X_1(t+1)$ 
18:          end if
19:        end if
20:      else
21:        { Narrowed exploration ( $X_2$ ) }
22:        Update the current solution using Equation (3).
23:        Apply one of the eight transfer functions on population  $X_2$ 
24:        if  $Fitness(X_2(t+1)) < Fitness(X(t))$  then
25:           $X(t) = X_2(t+1)$ 
26:          if  $Fitness(X_2(t+1)) < Fitness(X_{best}(t))$  then
27:            The  $X_{best}(t) = X_2(t+1)$ 
28:          end if
29:        end if
30:      end if
31:    else
32:      if  $rand \leq 0.5$  then
33:        { Expanded exploitation ( $X_3$ ) }
34:        Update the current solution using Equation (11).
35:        Apply one of the eight transfer functions on population  $X_3$ 
36:        if  $Fitness(X_3(t+1)) < Fitness(X(t))$  then
37:           $X(t) = X_3(t+1)$ 
38:          if  $Fitness(X_3(t+1)) < Fitness(X_{best}(t))$  then
39:            The  $X_{best}(t) = X_3(t+1)$ 
40:          end if
41:        end if
42:      else
43:        { Narrowed exploitation ( $X_4$ ) }
44:        Update the current solution using Equation (12).
45:        Apply one of the eight transfer functions on population  $X_4$ 
46:        if  $Fitness(X_4(t+1)) < Fitness(X(t))$  then
47:           $X(t) = X_4(t+1)$ 
48:          if  $Fitness(X_4(t+1)) < Fitness(X_{best}(t))$  then
49:            The  $X_{best}(t) = X_4(t+1)$ 
50:          end if
51:        end if
52:      end if
53:    end if
54:  end for
55: end while
56: The optimum solution ( $X_{best}$ ).

```

proven successful not only in low- and medium-sized knapsack datasets but also in large-scale knapsack datasets. The analyzes have been performed on MATLAB R2014a platform using a PC with Windows 10 platform, an Intel Core i5, 1.19 GHz CPU, and 12 GB of RAM.

3.1. Description of benchmark datasets

In this study, sixty-three 0–1 knapsack benchmark test datasets were selected to demonstrate the success of Binary AO (BAO). Table 2 shows the first 0–1 knapsack benchmark datasets, Table 3 shows the second 0–1 knapsack benchmark datasets, and Table 4 shows the third 0–1 knapsack benchmark datasets. BAO was run in these benchmark test datasets and its success was demonstrated. These datasets

were obtained from <https://pages.mtu.edu/~kreher/cages/Data.html>, http://artemisa.unicauca.edu.co/~johnyortega/instances_01_KP/, and (Kulkarni and Shabir, 2016) sources.

3.2. Parameter settings

Ten different values for population size were determined and population size was tested in detail at these values ($N = \{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$). The number of the maximum iteration was determined as 5000 ($T=5000$). Each study was evaluated as 20 independent studies. There are several reasons why the maximum number of iterations is fixed at the beginning of the algorithm. One of them is this. Heuristic algorithms obtain optimum results at a certain number of

Algorithm 3: Binary Aquila Optimizer Algorithm (BAO-CM) (with crossover and mutation)	
1:	Initialization the population X ($i=1, 2, 3, \dots, N$) (X = the current solution) (N =population size) (Dim =dimension)
2:	Initialize the parameters (i.e., N , Dim , S_{Dim} , T , etc).
3:	while ($t \leq T$) do
4:	Update the population X values using Equation (16).
5:	Calculate the fitness function values and determine the best-obtained solution ($X_{best}(t)$)
6:	for ($i=1, 2, \dots, N$) do
7:	if $t \leq T$ then
8:	{Crossover process} {Mutation process}
9:	Update the current solution using Equation (17).
10:	Apply the mutation operation on X_{new}
11:	if $Fitness(X_{new}(t+1)) < Fitness(X(t))$ then
12:	$X(t) = X_{new}(t+1)$
13:	if $Fitness(X_{new}(t+1)) < Fitness(X_{best}(t))$ then
14:	The $X_{best}(t) = X_{new}(t+1)$
15:	end if
16:	end if
17:	end if
18:	end for
19:	end while
20:	The optimum solution (X_{best}).

Table 2

0–1 knapsack benchmark datasets (first benchmarks).

Id	Problem	Capacity	Dimension	Optimum value
P1	Kp_1	269	10	295
P2	Kp_2	878	20	1024
P3	Kp_3	20	4	35
P4	Kp_4	11	4	23
P5	Kp_5	375	15	481.07
P6	Kp_6	60	10	52
P7	Kp_7	50	7	107
P8	Kp_8	10.000	23	9767
P9	Kp_9	80	5	130
P10	Kp_10	879	20	1025
P11	Kp_11	577	30	1437
P12	Kp_12	655	35	1689
P13	Kp_13	819	40	1821
P14	Kp_14	907	45	2033
P15	Kp_15	882	50	2440
P16	Kp_16	1050	55	2651
P17	Kp_17	1006	60	2917
P18	Kp_18	1319	65	2817
P19	Kp_19	1426	70	3223
P20	Kp_20	1433	75	3614

iterations. Sometimes this number of iterations can be very large. This increases the running time of the algorithm. This is usually undesirable. Instead, researchers show the maximum values they will achieve in a certain number of iterations (a reasonable run time) as a result. Another reason is that heuristic algorithms choose a fixed value for other parameter settings while determining various parameter settings such as population size. When the relevant literature studies were examined, it was seen that the heuristic algorithms they proposed mostly chose the maximum number of iterations as the stopping criterion. Each researcher should choose the stopping criterion equally when comparing his/her proposed algorithm with the literature fairly. In this study, the maximum number of iterations used by other comparison algorithms was chosen in order to make the literature comparisons fair. The reason why the number of runs is determined at the beginning of the algorithm is that heuristic algorithms can obtain a different value each time they run at certain parameter settings. They may not give an optimal result. In such cases, a single number of studies will not be enough to determine the success of the algorithm. Instead, an evaluation is made of the results by running the algorithm more than once. This situation is more effective in expressing the success of the algorithm. Related literature articles were examined and it was observed that 20 was preferred as the number of runs. In this study, the number of runs was chosen as 20 so that the algorithms are in equal race conditions.

Table 3

0–1 knapsack benchmark datasets (second benchmarks).

Id	Problem	Capacity	Dimension (Dim)	Optimum value
P21	Kp_8a	1863633	8	3924400
P22	Kp_8b	1822718	8	3813669
P23	Kp_8c	1609419	8	3347452
P24	Kp_8d	2112292	8	4187707
P25	Kp_8e	2493250	8	4955555
P26	Kp_12a	2805213	12	5688887
P27	Kp_12b	3259036	12	6498597
P28	Kp_12c	3489815	12	5170626
P29	Kp_12d	3453702	12	6992404
P30	Kp_12e	2520392	12	5337472
P31	Kp_16a	3780355	16	7850983
P32	Kp_16b	4426945	16	9352998
P33	Kp_16c	4323280	16	9151147
P34	Kp_16d	4550938	16	9348889
P35	Kp_16e	3760429	16	7769117
P36	Kp_20a	5169647	20	10727049
P37	Kp_20b	4681373	20	9818261
P38	Kp_20c	5063791	20	10714023
P39	Kp_20d	4286641	20	8929156
P40	Kp_20e	4476000	20	9357969
P41	Kp_24a	6404180	24	13549094
P42	Kp_24b	5971071	24	12233713
P43	Kp_24c	5870470	24	12448780
P44	Kp_24d	5762284	24	11815315
P45	Kp_24e	6654569	24	13940099

Table 4

Large-scale 0–1 knapsack benchmark datasets (third benchmarks).

Id	Problem	Capacity	Dimension	Optimum values
P46	Kp1_100	995	100	9147
P47	Kp1_200	1008	200	11238
P48	Kp1_500	2543	500	28857
P49	Kp1_1000	5002	1000	54503
P50	Kp1_2000	10011	2000	110625
P51	Kp1_5000	25016	5000	276457
P52	Kp2_100	995	100	1514
P53	Kp2_200	1008	200	1634
P54	Kp2_500	2543	500	4566
P55	Kp2_1000	5002	1000	9052
P56	Kp2_2000	10011	2000	18051
P57	Kp2_5000	25016	5000	44356
P58	Kp3_100	997	100	2397
P59	Kp3_200	997	200	2697
P60	Kp3_500	2517	500	7117
P61	Kp3_1000	4990	1000	14390
P62	Kp3_2000	9819	2000	28919
P63	Kp3_5000	24805	5000	72505

Table 5The mean results of BAO ($N=\{10, 20, 30, 40, 50, 60, 70, 80, 90, 100\}$; $T=\{5000\}$).

P_ID	P_Name	N=10	N=20	N=30	N=40	N=50	N=60	N=70	N=80	N=90	N=100
P21	Kp.8a	3924400	3924400	3924400	3924400	3924400	3924400	3924400	3924400	3924400	3924400
P22	Kp.8b	3813669	3813669	3813669	3813669	3813669	3813669	3813669	3813669	3813669	3813669
P23	Kp.8c	3347452	3347452	3347452	3347452	3347452	3347452	3347452	3347452	3347452	3347452
P24	Kp.8d	4187707	4187707	4187707	4187707	4187707	4187707	4187707	4187707	4187707	4187707
P25	Kp.8e	4955555	4955555	4955555	4955555	4955555	4955555	4955555	4955555	4955555	4955555

Table 6The mean results of the BAO-CM ($S_{Dim}=\{0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$; $T=5000$; $N=20$).

P_ID	P_Name	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
P1	Kp.1	295	295	295	295	295	295	295	295	295
P2	Kp.2	1024	1024	1024	1024	1024	1024	1024	1024	1024
P3	Kp.3	33	33	33	33	33	33	33	33	35
P4	Kp.4	23	23	23	23	23	23	23	23	23
P5	Kp.5	481.07	481.07	481.07	481.07	481.07	481.07	481.07	481.07	481.07
P6	Kp.6	52	52	52	52	52	52	52	52	52
P7	Kp.7	107	107	107	107	107	107	107	107	107
P8	Kp.8	9757.25	9756.50	9757.55	9758.70	9758.80	9761.15	9756.90	9760.05	9767
P9	Kp.9	130	130	130	130	130	130	130	130	130
P10	Kp.10	1025	1025	1025	1025	1025	1025	1025	1025	1025
P11	Kp.11	1414.15	1417.10	1420.05	1420.30	1424.10	1424.10	1420.05	1427.90	1437
P12	Kp.12	1667.15	1667.60	1672.65	1672.00	1678.65	1677.20	1678.75	1679.60	1689
P13	Kp.13	1516.85	1533.50	1532.80	1541.05	1548.80	1546.45	1555.15	1556.95	1821
P14	Kp.14	1946.90	1953.20	1950.25	1960.30	1966.00	1969.80	1968.90	1970.35	2033
P15	Kp.15	2338.60	2354.65	2346.05	2357.80	2364.45	2366.60	2370.05	2374.60	2444
P16	Kp.16	2512.50	2507.35	2530.50	2525.65	2535.20	2536.50	2543.60	2543.60	2651
P17	Kp.17	2768.15	2766.20	2777.60	2765.70	2773.70	2785.35	2797.00	2779.75	2917
P18	Kp.18	2661.75	2662.00	2678.40	2675.50	2683.25	2681.20	2685.20	2694.30	2818
P19	Kp.19	3063.35	3064.15	3076.35	3076.50	3083.95	3095.80	3102.60	3087.75	3161.00
P20	Kp.20	3367.05	3370.95	3367.95	3395.75	3393.00	3393.00	3408.80	3417.65	3614

***Bold** values indicate the optimal results.

Table 5 shows the success of ten different population values on five different knapsack datasets. The results showed that the population value did not have a significant effect on the knapsack problem. The population value was chosen as 20 to make a fair comparison.

Other constant parameters affecting the success of AO are α and δ . A detailed study was made for these constant values in the original paper on AO, and the most appropriate values were determined as 0.1 and 0.1, respectively (Abualigah et al., 2021). It was used in the same way in this study.

Another parameter that affects the success of BAO is S_{Dim} . S_{Dim} shows the number of dimensions selected in the crossover process. Performance tests were performed on twenty different knapsack problems for nine different S_{Dim} values {0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, and 0.9}. The results are shown in Table 6. According to the results, the most appropriate value was determined as 0.9.

Table 7 shows the parameters set up for BAO.

In this study, the performance of each algorithm was evaluated using five criteria. These criteria are best, worst, mean, standard deviation (Std), and gap. The gap is a statistical test used to check how close each algorithm is to the optimal solution (Opt) and is calculated using the following formula:

$$Gap = \frac{Optimum\ value - Mean\ value}{Optimum\ value} \times 100 \text{ for Maximization problem} \quad (20)$$

Heuristic algorithms do not guarantee the same result every time they run. Therefore, they are run for a certain number of runs. Statistical calculations are extremely useful in evaluating the results. It gives us information about the performance of the proposed algorithms.

3.3. Choosing the most successful transfer function on BAO

Eight different transfer functions were chosen to convert the continuous search space to binary search space in BAO. Which of these transfer functions gives more successful results is tested in detail in this

Table 7

The parameter settings of BAO.

Parameters	Values
Population size (N)	20
Number of runs	20
Maximum iterations (T)	5000
α and δ	0.1 and 0.1
S_{Dim}	0.9

subsection. Results were determined according to the best, worst, mean, standard deviation, and gap comparison criteria. In all BAO variations, the maximum iteration was chosen as 500 and the population amount was chosen as 20. The number of independent runs was chosen as twenty. BAO variations were tested on twenty low-scale knapsack problems.

In Table 8, BAO variations are compared according to the best value. According to the results, BAO1 showed superior success in 19 out of 20 knapsack problems. BAO2, BAO4, and BAO5 showed superior success in 10 of 20 knapsack problems. Other BAO variations were less than 50 percent successful. Table 9 shows the worst results of BAO variations.

Mean results for eight different variations of BAO are shown in Table 10. According to the results, BAO1 showed superior success in 17 of 20 knapsack problems. BAO2 and BAO4 showed superior success in 8 of 20 knapsack problems. Other BAO variations were successful in only 8 of the 20 knapsack problems.

Standard deviation results for eight different variations of BAO are shown in Table 11. According to the results, BAO1 showed superior success in 14 of 20 knapsack problems. BAO2 and BAO4 showed superior success in 9 of 20 knapsack problems. Other BAO variations were successful in 6 of the 20 knapsack problems. Gap results for eight different variations of BAO are shown in Table 12. The results of the gap and mean BAO variations were similarly successful.

Fig. 11 shows the comparison of the fitness values for the variations of the BAO. According to the results, the most successful BAO variation was BAO1, and the BAO1 variation was used in other comparisons as BAO-T.

Table 8

The best of the BAO results for eight different transfer functions.

Id	Name	Opt	BAO1	BAO2	BAO3	BAO4	BAO5	BAO6	BAO7	BAO8
P1	Kp_1	295	295	295	295	295	295	295	295	295
P2	Kp_2	1024	1024	1024	1024	1024	1024	1024	1004	984
P3	Kp_3	35	35	35	35	35	35	35	35	35
P4	Kp_4	23	23	23	23	23	23	23	23	23
P5	Kp_5	481.07	481.07	481.07	481.07	481.07	481.07	481.07	481.07	481.07
P6	Kp_6	52	52	52	52	52	52	52	52	52
P7	Kp_7	107	107	107	107	107	107	107	107	107
P8	Kp_8	9767	9767	9765	9765	9767	9767	9765	9763	9758
P9	Kp_9	130	130	130	130	130	130	130	130	130
P10	Kp_10	1025	1025	1025	1025	1025	1025	1019	992	979
P11	Kp_11	1437	1425	1418	1403	1398	1392	1338	1300	1314
P12	Kp_12	1689.0	1679	1665	1647	1632	1630	1593	1538	1450
P13	Kp_13	1821	1587	1581	1579	1544	1536	1503	1383	1357
P14	Kp_14	2033	1992	1955	1923	1864	1811	1760	1696	1709
P15	Kp_15	2440	2366	2337	2236	2255	2166	2133	2062	1924
P16	Kp_16	2651	2554	2488	2480	2403	2308	2287	2235	2179
P17	Kp_17	2917	2808	2710	2673	2643	2543	2571	2306	2218
P18	Kp_18	2817	2693	2698	2570	2514	2488	2368	2232	2169
P19	Kp_19	3223	3131	3062	2898	2934	2756	2737	2714	2448
P20	Kp_20	3614	3433	3308	3265	3194	3099	3002	2951	2818

***Bold** values indicate the optimal results.**Table 9**

The worst of the BAO results for eight different transfer functions.

Id	Name	Opt	BAO1	BAO2	BAO3	BAO4	BAO5	BAO6	BAO7	BAO8
P1	Kp_1	295	295	295	295	295	295	295	295	295
P2	Kp_2	1024	1024	1018	1016	996	978	964	907	886
P3	Kp_3	35	35	35	35	35	35	35	35	35
P4	Kp_4	23	23	23	23	23	23	23	23	23
P5	Kp_5	481.07	475.48	481.07	475.48	481.07	475.48	469.16	454.43	429.28
P6	Kp_6	52	52	52	52	52	52	52	52	52
P7	Kp_7	107	107	107	107	107	107	107	107	107
P8	Kp_8	9767	9750	9751	9757.00	9758	9752	9748	9734	9707
P9	Kp_9	130	130	130	130	130	130	130	130	130
P10	Kp_10	1025	1025	1019	1014	1005	983	970	915	902
P11	Kp_11	1437	1405	1379	1347	1335	1295	1245	1206	1119
P12	Kp_12	1689.0	1653	1604	1568	1558	1457	1454	1382	1266
P13	Kp_13	1821	1478	1505	1474	1491	1386	1334	1257	1169
P14	Kp_14	2033	1903	1883	1811	1789	1701	1665	1554	1462
P15	Kp_15	2440	2292	2230	2144	2110	1985	1969	1827	1728
P16	Kp_16	2651	2453	2414	2322	2313	2173	2104	1963	1858
P17	Kp_17	2917	2701	2608	2541	2484	2306	2308	2134	2035
P18	Kp_18	2817	2622	2548	2436	2408	2272	2193	2051	1950
P19	Kp_19	3223	3008	2896	2785	2724	2608	2495	2530	2189
P20	Kp_20	3614	3322	3224	3044	2994	2841	2748	2733	2425

Table 10

The mean of the BAO results for eight different transfer functions.

Id	Name	Opt	BAO1	BAO2	BAO3	BAO4	BAO5	BAO6	BAO7	BAO8
P1	Kp_1	295	295	295	295	295	295	295	295	295
P2	Kp_2	1024	1024	1023.7	1019.7	1014.8	1009.8	1000.85	956.45	932.1
P3	Kp_3	35	35	35	35	35	35	35	35	35
P4	Kp_4	23	23	23	23	23	23	23	23	23
P5	Kp_5	481.07	480.23	481.07	480.79	481.07	480.79	479.91	475.96	459.97
P6	Kp_6	52	52	52	52	52	52	52	52	52
P7	Kp_7	107	107	107	107	107	107	107	107	107
P8	Kp_8	9767	9756.70	9758.05	9760.25	9760.75	9760.25	9754.85	9747.85	9735.35
P9	Kp_9	130	130	130	130	130	130	130	130	130
P10	Kp_10	1025	1025	1024.70	1021.60	1017.3	1010.1	995.15	960.15	942.45
P11	Kp_11	1437	1414	1399.75	1370.70	1366.35	1338.2	1300.15	1249.6	1214.75
P12	Kp_12	1689.0	1665.85	1633.60	1608.60	1581.95	1536.15	1499	1448.35	1371.45
P13	Kp_13	1821	1517.75	1532.15	1514.55	1511.4	1445.2	1412.35	1317.95	1261.95
P14	Kp_14	2033	1947.50	1910.25	1847.75	1825.25	1759.75	1706.5	1610.95	1563.3
P15	Kp_15	2440	2331.20	2265.70	2191.20	2162.25	2076.4	2024.3	1940.3	1808.85
P16	Kp_16	2651	2503.10	2444.90	2386.15	2350.55	2250.55	2201.75	2080.45	1967.75
P17	Kp_17	2917	2749.80	2657.70	2582.60	2535.05	2430.4	2369.75	2231.5	2115.35
P18	Kp_18	2817	2650.35	2599.70	2503.50	2468.85	2355.3	2279.4	2155.7	2043.4
P19	Kp_19	3223	3055.60	2953.80	2843.45	2809.9	2667.65	2609.6	2618.75	2301.95
P20	Kp_20	3614	3358.75	3262.10	3143.05	3083.65	2946.2	2865.75	2851.15	2525.7

***Bold** values indicate the optimal results.

Table 11

The standard deviation of the BAO results for eight different transfer functions.

Id	Name	Opt	BAO1	BAO2	BAO3	BAO4	BAO5	BAO6	BAO7	BAO8
P1	Kp.1	295	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P2	Kp.2	1024	0.00	1.31	3.24	7.59	12.40	18.07	22.28	25.73
P3	Kp.3	35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P4	Kp.4	23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P5	Kp.5	481.07	2.00	0.00	1.22	0.00	1.22	2.98	6.75	14.23
P6	Kp.6	52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P7	Kp.7	107	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P8	Kp.8	9767	3.61	3.40	2.36	2.12	4.11	4.66	7.43	13.28
P9	Kp.9	130	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P10	Kp.10	1025	0.00	1.31	3.99	6.56	12.43	14.85	18.60	22.75
P11	Kp.11	1437	5.01	12.40	13.23	16.75	21.64	24.89	27.93	45.96
P12	Kp.12	1689.0	6.56	15.72	23.72	20.12	40.34	31.72	45.59	46.68
P13	Kp.13	1821	23.40	21.06	24.28	14.03	37.92	51.62	36.56	49.19
P14	Kp.14	2033	20.11	19.31	28.89	18.65	29.51	25.29	36.23	56.51
P15	Kp.15	2440	19.53	23.94	26.67	33.82	47.11	37.33	58.52	45.20
P16	Kp.16	2651	22.07	18.75	32.92	22.85	40.11	54.69	59.74	63.33
P17	Kp.17	2917	23.28	32.28	38.25	37.33	66.49	54.96	45.25	49.11
P18	Kp.18	2817	21.68	39.33	34.14	29.70	59.54	49.25	45.23	55.55
P19	Kp.19	3223	34.15	38.69	33.14	51.82	45.74	55.64	58.35	60.27
P20	Kp.20	3614	29.42	24.08	46.99	54.97	77.03	60.11	50.81	82.75

*Bold values indicate the optimal results.

Table 12

The gap results of the BAO for eight different transfer functions.

Id	Name	Opt	BAO1	BAO2	BAO3	BAO4	BAO5	BAO6	BAO7	BAO8
P1	Kp.1	295	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P2	Kp.2	1024	0.00	0.03	0.42	0.90	1.39	2.26	6.60	8.97
P3	Kp.3	35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P4	Kp.4	23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P5	Kp.5	481.07	0.17	0.00	0.06	0.00	0.06	0.24	1.06	4.39
P6	Kp.6	52	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P7	Kp.7	107	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P8	Kp.8	9767	0.11	0.09	0.07	0.06	0.07	0.12	0.20	0.32
P9	Kp.9	130	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P10	Kp.10	1025	0.00	0.03	0.33	0.75	1.45	2.91	6.33	8.05
P11	Kp.11	1437	1.60	2.59	4.61	4.92	6.88	9.52	13.04	15.47
P12	Kp.12	1689.0	1.37	3.28	4.76	6.34	9.05	11.25	14.25	18.80
P13	Kp.13	1821	16.65	15.86	16.83	17.00	20.64	22.44	27.62	30.70
P14	Kp.14	2033	4.21	6.04	9.11	10.22	13.44	16.06	20.76	23.10
P15	Kp.15	2440	4.46	7.14	10.20	11.38	14.90	17.04	20.48	25.87
P16	Kp.16	2651	5.58	7.77	9.99	11.33	15.11	16.95	21.52	25.77
P17	Kp.17	2917	5.73	8.89	11.46	13.09	16.68	18.76	23.50	27.48
P18	Kp.18	2817	5.92	7.71	11.13	12.36	16.39	19.08	23.48	27.46
P19	Kp.19	3223	5.19	8.35	11.78	12.82	17.23	19.03	18.75	28.58
P20	Kp.20	3614	7.32	9.99	13.27	14.91	18.70	20.92	21.33	30.31

*Bold values indicate the optimal results.

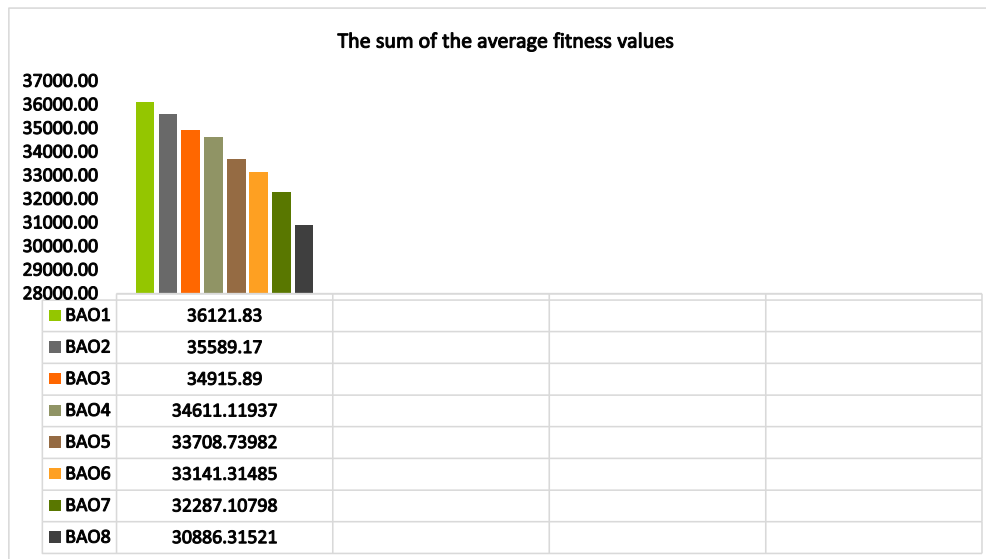
**Fig. 11.** Comparison of the fitness values for the variations of the BAO.

Table 13

The BAO-T and BAO-CM results for the first group of knapsack problems.

Id	Name	Opt	BAO-T				BAO-CM			
			Best	Mean	Std	Gap	Best	Mean	Std	Gap
P1	Kp_1	295	295	295	0.00	0.00	295	295	0.00	0.00
P2	Kp_2	1024	1024	1024	0.00	0.00	1024	1024	0.00	0.00
P3	Kp_3	35	35	35	0.00	0.00	35	35	0.00	0.00
P4	Kp_4	23	23	23	0.00	0.00	23	23	0.00	0.00
P5	Kp_5	481.07	481.07	480.23	2.00	0.17	481.07	481.07	0.00	0.00
P6	Kp_6	52	52	52	0.00	0.00	52	52	0.00	0.00
P7	Kp_7	107	107	107	0.00	0.00	107	107	0.00	0.00
P8	Kp_8	9767	9767	9756.70	3.61	0.11	9767	9767	0.00	0.00
P9	Kp_9	130	130	130	0.00	0.00	130	130	0.00	0.00
P10	Kp_10	1025	1025	1025	0.00	0.00	1025	1025	0.00	0.00
P11	Kp_11	1437	1425	1414	5.01	1.60	1437	1437	0.00	0.00
P12	Kp_12	1689.0	1679	1665.85	6.56	1.37	1689	1689	0.00	0.00
P13	Kp_13	1821	1587	1517.75	23.40	16.65	1821	1821	0.00	0.00
P14	Kp_14	2033	1992	1947.50	20.11	4.21	2033	2033	0.00	0.00
P15	Kp_15	2440	2366	2331.20	19.53	4.46	2444	2444	0.00	0.00
P16	Kp_16	2651	2554	2503.10	22.07	5.58	2651	2651	0.00	0.00
P17	Kp_17	2917	2808	2749.80	23.28	5.73	2917	2917	0.00	0.00
P18	Kp_18	2817	2693	2650.35	21.68	5.92	2818	2818	0.00	0.00
P19	Kp_19	3223	3131	3055.60	34.15	5.19	3180	3161.00	4.36	1.92
P20	Kp_20	3614	3433	3358.75	29.42	7.32	3614	3614	0.00	0.00

***Bold** values indicate the optimal results.**Table 14**

The BAO-T and BAO-CM results for the second group knapsack problems.

Id	Name	Opt	BAO-T				BAO-CM			
			Best	Mean	Std	Gap	Best	Mean	Std	Gap
P21	Kp_8a	3924400	3924400	3924400	0.00	0.00	3924400	3924400	0.00	0.00
P22	Kp_8b	3813669	3813669	3813669	0.00	0.00	3813669	3813669	0.00	0.00
P23	Kp_8c	3347452	3347452	3347452	0.00	0.00	3347452	3347452	0.00	0.00
P24	Kp_8d	4187707	4187707	4187707	0.00	0.00	4187707	4187707	0.00	0.00
P25	Kp_8e	4955555	4955555	4955555	0.00	0.00	4955555	4955555	0.00	0.00
P26	Kp_12a	5688887	5688887	5688887	0.00	0.00	5688887	5688887	0.00	0.00
P27	Kp_12b	6473019	6498597	6498597	0.00	0.00	6498597	6498597	0.00	0.00
P28	Kp_12c	5170626	5170626	5170626	0.00	0.00	5170626	5170626	0.00	0.00
P29	Kp_12d	6941564	6992404	6992404	0.00	0.00	6992404	6992404	0.00	0.00
P30	Kp_12e	5337472	5337472	5320706	22847.8	0.31	5337472	5337472	0.00	0.00
P31	Kp_16a	7850983	7850983	7822759	17974.75	0.36	7850983	7850983	0.00	0.00
P32	Kp_16b	9352998	9352998	9299500	44678.91	0.57	9352998	9352998	0.00	0.00
P33	Kp_16c	9151147	9151147	9119861	28382.66	0.34	9151147	9151147	0.00	0.00
P34	Kp_16d	9348889	9348889	9326542	21850.95	0.24	9348889	9348889	0.00	0.00
P35	Kp_16e	7769117	7769117	7759291	9250.46	0.13	7769117	7769117	0.00	0.00
P36	Kp_20a	10727049	10717569	10631817	40736.37	0.89	10727049	10727049	0.00	0.00
P37	Kp_20b	9818261	9818261	9740051	31356.65	0.80	9818261	9818261	0.00	0.00
P38	Kp_20c	10714023	10714023	10623099	58919.13	0.85	10714023	10714023	0.00	0.00
P39	Kp_20d	8929156	8889920	8827026	36907.38	1.14	8929156	8929156	0.00	0.00
P40	Kp_20e	9357969	9357192	9290988	35604.71	0.72	9357969	9357969	0.00	0.00
P41	Kp_24a	13549094	13458609	13363882	43821.19	1.37	13549094	13549094	0.00	0.00
P42	Kp_40b	12233713	12154685	12071440	37218.2	1.33	12233713	12233713	0.00	0.00
P43	Kp_24c	12448780	12350058	12280388	29766.15	1.35	12448780	12448780	0.00	0.00
P44	Kp_24d	11815315	11754633	11665735	45549.72	1.27	11815109.72	11815109.72	0.00	0.0017
P45	Kp_24e	13940099	13850417	13759686	44737.25	1.29	13940099	13940099	0.00	0.00

***Bold** values indicate the optimal results.

3.4. BAO-T and BAO-CM results in small and middle-dimensional knapsack problem datasets

In this subsection, the performance results of BOA-T in second group knapsack problems are obtained. In addition, the performance results of the BAO-CM algorithm proposed in this section are obtained in the first and second groups' knapsack problems. The best, mean, standard deviation (Std), and gap values of the obtained BAO-T and BAO-CM results were compared. BAO-T and BAO-CM results for the first and second groups' knapsack problems are shown in [Tables 13 and 14](#). According to the results, BAO-CM achieved better results than BAO-T in knapsack problems. BAO-CM got this success from crossover and mutation operators. BAO-CM achieved optimum results in 43 of the first and second group knapsack problems. BAO-T achieved optimum results in 17 of the first and second group knapsack problems.

3.5. Comparisons of BAO-T and BAO-CM with the different algorithms

3.5.1. Comparisons of BAO-T and BAO-CM with the different algorithms for the first group of the benchmark dataset

In this subsection, BAO-T and BAO-CM algorithms are compared with various heuristic algorithms developed in recent years for first knapsack benchmark datasets. These algorithms are a Binary Equilibrium Optimization Algorithm ([Abdel-Basset et al., 2021](#)) (BEO), a Whale Optimization algorithm ([Mirjalili and Lewis, 2016](#)) (WOS), a Flower Pollination algorithm ([Yang et al., 2014](#)) (FPS), a Bat Algorithm ([Yang, 2010](#)) (BAS), a Crow Search Algorithm ([Askarzadeh, 2016](#)) (CSAS), a Particle Swarm Optimization algorithm ([Eberhart and Kennedy, 1995](#)) (PSOS), a Complex-valued encoding Wind-Driven Optimization ([Zhou et al., 2017](#)) (CWDO), and a Discrete Improved Artificial Bee Colony algorithm ([Zhang and Liu, 2019](#)) (DIABC). Comparison

Table 15

The comparisons of BAO-T and BAO-CM with other algorithms (according to mean results).

Id	BEO	WOS	PSOS	FPS	CSAS	BAS	CWDO	DIABC	BAO-T	BAO-CM
P1	295	295	295	295	295	295	295	295	295	295
P2	1024	1024	1024	1024	1024	1024	1024	1024	1024	1024
P3	35	35	35	35	35	35	35	35	35	35
P4	23	23	23	23	23	23	23	23	23	23
P5	481.07	481.07	481.07	481.07	481.07	481.07	481.07	481.07	480.23	481.07
P6	52	52	52	52	52	52	52	52	52	52
P7	107	107	107	107	107	107	107	107	107	107
P8	9767	9763.17	9766.55	9766.55	9767	9767	9753	9753	9756.70	9767
P9	130	130	130	130	130	130	130	130	130	130
P10	1025	1025	1025	1025	1025	1025	1025	1025	1025	1025
P11	1437	1437	1437	1437	1437	1437	–	–	1414	1437
P12	1689	1687.9	1688.2	1689	1689	1689	–	–	1665.85	1689
P13	1821	1818.8	1821	1821	1821	1821	–	–	1517.75	1821
P14	2033	2033	2033	2033	2033	2033	–	–	1947.50	2033
P15	2444	2440.1	2442.15	2441.3	2442.4	2441.3	–	–	2331.20	2444
P16	2651	2651	2648.15	2643.8	2651	2650	–	–	2503.10	2651
P17	2917	2917	2917	2911	2913.2	2917	–	–	2749.80	2917
P18	2818	2815.95	2809.0	2797.8	2794.25	2809.5	–	–	2650.35	2818
P19	3223	3220.55	3208.55	3186	3220.25	3215.8	–	–	3055.60	3161.00
P20	3614	3600.3	3592	3562.4	3559.95	3588.24	–	–	3358.75	3614

***Bold** values indicate the optimal results.

Table 16

The comparisons of BAO-T and BAO-CM with other algorithms (according to standard deviation results).

Id	BEO	WOS	PSOS	FPS	GSAS	BAS	CWDO	DIABC	BAO-T	BAO-CM
P1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P5	0.00	0.00	0.00	0.00	0.00	0.00	2.9E–13	0.00	2.00	0.00
P6	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P8	0.00	5.58	2.191	2.191	0.00	0.00	3.5	3.5	3.61	0.00
P9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P11	0.00	0.00	0.00	0.00	0.00	0.00	–	–	5.01	0.00
P12	0.00	2.62	2.874	0.00	0.00	0.00	–	–	6.56	0.00
P13	0.00	2.003	0.00	0.00	0.00	0.00	–	–	23.40	0.00
P14	0.00	0.00	0.00	0.00	0.00	0.00	–	–	20.11	0.00
P15	0.00	3.044	3.072	3.004	2.013	4.618	–	–	19.53	0.00
P16	0.00	0.00	3.940	3.92	0.00	4.35	–	–	22.07	0.00
P17	0.00	0.00	0.00	2911	2913.2	0.00	–	–	23.28	0.00
P18	0.00	1.533	14.953	25.22	12.2	35.56	–	–	21.68	0.00
P19	0.00	1.59	17.35	23.11	0.52	10.93	–	–	34.15	4.36
P20	0.00	7.217	25.683	48.53	16.59	46.35	–	–	29.42	0.00

***Bold** values indicate the optimal results.

results are shown in Tables 15–17. Figs. 12 and 13 show the comparisons of the gap values of the datasets for the first benchmarks. According to the results, the algorithm that reached the optimum results for the 20 knapsack dataset was BEO. On the other hand, BAO-CM obtained optimum results for 19 datasets after BEO.

3.5.2. Comparison of BAO-T and BAO-CM with the different algorithms for the second group of the benchmark dataset

Comparison 1: In this subsection, BAO-T and BAO-CM algorithms are compared with various heuristic algorithms developed in recent years for second knapsack benchmark datasets. These algorithms are a Binary Equilibrium Optimization Algorithm (Abdel-Basset et al., 2021) (BEO), a Whale Optimization algorithm (Mirjalili and Lewis, 2016) (WOS), a Flower Pollination algorithm (Yang et al., 2014) (FPS), a Bat Algorithm (Yang, 2010) (BAS), a Crow Search Algorithm (Askarzadeh, 2016) (CSAS), a Particle Swarm Optimization algorithm (Eberhart and Kennedy, 1995) (PSOS), an Improved Monkey Algorithm (Zhou et al., 2016a) (CGMA), and a Hybrid Symbiotic Organism Search algorithm

(Wu et al., 2018) (GHSOS). Comparison results are shown in Tables 18–20. Table 21 shows the numbers of wins, draws, and losses to show one-to-one superiority or inadequacy between the proposed method and other binary variables. If the proposed method has a better void result than the other algorithm, ‘+’ is used, ‘–’ offers worse and the same performance is offered with ‘=’. Algorithms were compared to three different criteria. These are the mean, standard deviation (Std), and gap values. Fig. 14 shows the comparisons of the average of the gap values of the datasets for the second benchmark.

According to the mean, standard deviation, and gap of the results, BAO-CM and BEO were the algorithms that showed superior success for 24 out of 25 knapsack datasets. According to Fig. 14, the most successful average gap results belong to BAO-CM. This shows that BAO-CM is the most successful algorithm overall in the second benchmark dataset.

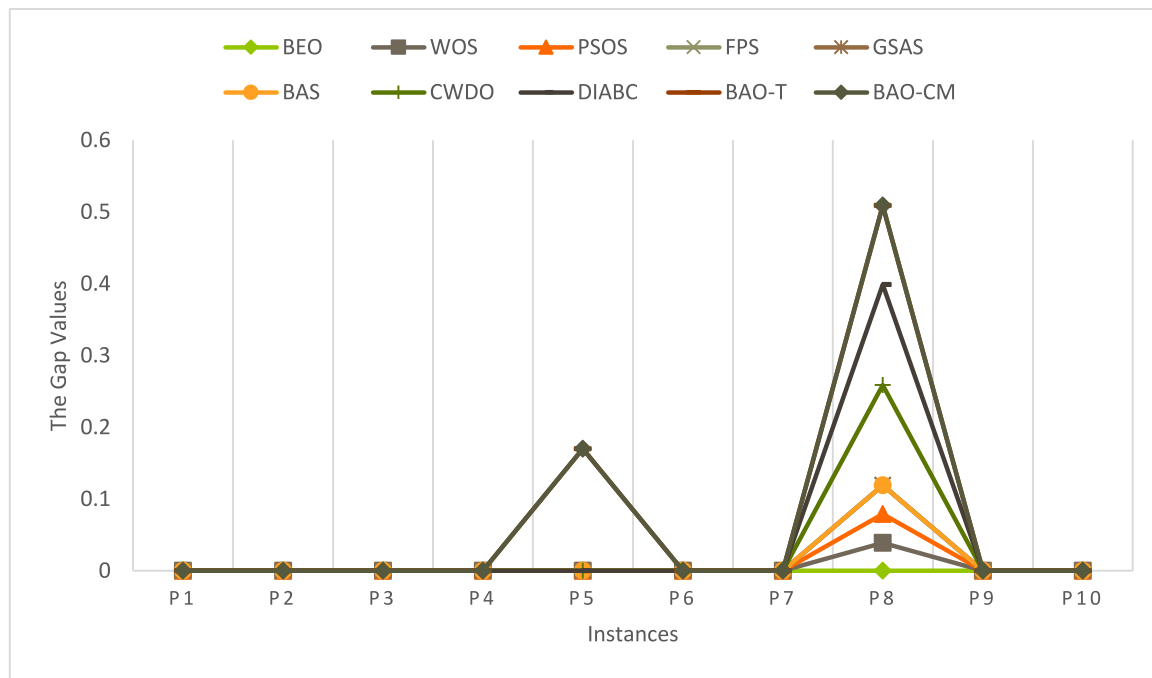
Comparison 2: In this subsection, BAO-T and BAO-CM algorithms are compared with various binary variations of the heuristic algorithms developed in recent years for second knapsack benchmark datasets. These binary variants are BPSO (Kennedy and Eberhart, 1997), MBPSO

Table 17

The comparisons of BAO-T and BAO-CM with other algorithms (according to gap results).

Id	BEO	WOS	PSOS	FPS	GSAS	BAS	CWDO	DIABC	BAO-T	BAO-CM
P1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P3	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P4	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P5	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.17	0.00
P6	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P8	0.00	0.039	0.040	0.040	0.00	0.00	0.14	0.14	0.11	0.00
P9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P11	0.00	0.00	0.00	0.00	0.00	0.00	–	–	1.60	0.00
P12	0.00	0.065	0.047	0.00	0.00	0.00	–	–	1.37	0.00
P13	0.00	0.1208	0.00	0.00	0.00	0.00	–	–	16.65	0.00
P14	0.00	0.00	0.00	0.00	0.00	0.00	–	–	4.21	0.00
P15	0.00	0.159	0.0756	0.110	0.065	0.110	–	–	4.46	0.00
P16	0.00	0.00	0.10	0.27	0.00	0.03	–	–	5.58	0.00
P17	0.00	0.00	0.00	0.20	0.13	0.00	–	–	5.73	0.00
P18	0.00	0.072	0.28	0.682	0.80	0.266	–	–	5.92	0.00
P19	0.00	0.076	0.448	1.147	0.085	0.22	–	–	5.19	1.92
P20	0.00	0.379	0.608	1.427	1.49	0.712	–	–	7.32	0.00

*Bold values indicate the optimal results.

**Fig. 12.** Comparisons of the gap values of the datasets from P1 to P10.

(Bansal and Deep, 2012), NGHS (Zou et al., 2011), DGHS (Xiang et al., 2014), S-bAFSA (Azad et al., 2014), CGMA (Zhou et al., 2016b), and BinEHO (Hakl, 2020). For fair comparisons, the maximum iteration value is set to 1000 for P21–P30 and 5000 for P31–P45. Experiment results are taken directly from Hakl (2020). Comparison results are shown in Tables 22 and 23. Algorithms were compared to two different criteria. These are the mean and gap values. Table 24 shows the numbers of wins, draws, and losses to show one-to-one superiority or inadequacy between the proposed method and other binary variables. If the proposed method has a better void result than the other algorithm, ‘+’ is used, ‘–’ offers worse and the same performance is offered with ‘=’. Fig. 15 shows the comparisons of the average of the gap values of the datasets for the second benchmark.

According to the mean and gap of the results, BAO-CM was the algorithm that showed superior success for 25 out of 25 knapsack datasets. After BAO-CM, the best success has been BinEHO and CGMA. The worst results belong to the BPSO and MBPSO algorithms. According

to Fig. 15, the most successful average gap results belong to BAO-CM. This shows that BAO-CM is the most successful algorithm overall in the second benchmark dataset.

3.6. Comparisons of BAO-CM for large-scale knapsack problems (for third group benchmarks)

In this subsection, eighteen large-scale 0–1 knapsack benchmark test datasets were selected to demonstrate the success of Binary AO. BAO was run in these benchmark test datasets and its success was demonstrated on large-scale datasets. This dataset consists of large dimensions from 100 to 5000. In addition, the success of BAO has been compared with the heuristic algorithms developed in the literature in recent years. There are the Genetic Algorithm (Ezugwu et al., 2019) (GA) Simulated Annealing (Ezugwu et al., 2019) (SA), Greedy Search Algorithm (Ezugwu et al., 2019) (GSA), hybrid GA-SA (Ezugwu et al., 2019), Branch and Bound algorithm (Ezugwu et al., 2019) (BB), and



Fig. 13. Comparisons of the gap values of the datasets from P11 to P20.

Table 18

The comparisons of BAO-T and BAO-CM with other algorithms (according to mean results).

Id	BEO	WOS	PSOS	FPS	CSAS	BAS	CGMA	GHSOS	BAO-T	BAO-CM
P21	3924400	3924400	3924400	3924400	3924400	3924400	3924400	3924400	3924400	3924400
P22	3813669	3805921	3813669	3813669	3813669	3813669	3813669	3813669	3813669	3813669
P23	3347452	3347452	3347452	3347452	3347452	3347452	3347452	3347452	3347452	3347452
P24	4187707	4187707	4187707	4187707	4187707	4187707	4187707	4187707	4187707	4187707
P25	4955555	4955555	4955555	4955555	4955555	4955555	–	–	4955555	4955555
P26	5688887	5683266	5688887	5688887	5688887	5688887	5683266	5688887	5688887	5688887
P27	6498597	6494760	6498597	6498597	6488365	6498597	6494760	6498597	6498597	6498597
P28	5170626	5170626	5170626	5170626	5170626	5170626	5170626	5170626	5170626	5170626
P29	6941564	6992404	6992404	6992404	6971781	6992404	6992404	6992404	6992404	6992404
P30	5337472	5337472	5337472	5337472	5337472	5337472	–	5337472	5320706	5337472
P31	7850983	7832971	7850983	7850983	7850983	7850983	7831152.5	7850983	7822759	7850983
P32	9352998	9334569.85	9352998	9352998	9352998	9352998	334569.85	9352998	9299500	9352998
P33	9151147	9151147	9151147	9151147	9151147	9151147	9151147	9151147	9119861	9151147
P34	9348889	9336691	9348889	9348889	9348889	9347059.3	9345229.2	9348889	9326542	9348889
P35	7769117	7769117	7761876.6	7769117	7769117	7769117	7769117	7769117	7759291	7769117
P36	10727049	10727049	7071775.5	10727049	10727049	10727049	10727049	10727049	10631817	10727049
P37	9818261	9818261	9818261	9818261	9818261	9818261	9818261	9818261	9740051	9818261
P38	10714023	10707307.9	10714023	10714023	10714023	10713732.8	10712553.7	10714023	10623099	10714023
P39	8929156	8917460	925028	8929156	8929156	8929156	8918423.2	8929156	8827026	8929156
P40	9357969	9352636.4	9357969	9357969	9357969	9357969	9357922.38	9357456.18	9290988	9357969
P41	13549094	13513046.15	13549094	13549094	13549094	13549094	13538097.1	13546441.04	13363882	13549094
P42	12233713	12233713	12233713	12233713	12233713	12233713	12233713	12233713	12071440	12233713
P43	12448780	12448780	12448780	12448780	12448780	12448780	12448618.3	12448780	12280388	12448780
P44	11814367	11810051	11810840.6	11810840.6	11810840.6	11811103.8	11808614.9	11815209.72	11665735	11815109.72
P45	13940099	13931016.65	13940099	13940099	13940099	13940099	13940099	13940099	13759686	13940099

*Bold values indicate the optimal results.

Binary Equilibrium Optimization Algorithm (Abdel-Basset et al., 2021) (EOS2). Experiment results are taken directly from Haklı (Abdel-Basset et al., 2021). Comparisons of the mean results are shown in Table 25 and comparisons of the gap results are shown in Table 26. The comparisons of BAO-CM with other algorithms on the large-scale datasets (according to one to one) are shown in Table 27. Fig. 16 shows the comparison of the gap values of the datasets from P46 to P63.

According to the mean and gap results, BAO-CM produced better results than other heuristic algorithms in 17 of 18 datasets. EOS2 showed superior performance in 13 of 18 datasets. After BAO-CM, the most successful algorithm was EOS2. Other algorithms performed less well on large-scale knapsack problems.

4. Conclusion

Meta-heuristic algorithms are often used because of their superior performance in solving NP-hard problems. Various heuristic algorithms are available in the literature (such as Monarch Butterfly Optimization (MBO), Earthworm Optimization Algorithm (EWA), Elephant Herding Optimization (EHO), Moth Search (MS) algorithm, Slime Mould Algorithm (SMA), Hunger Games Search (HGS), Runge Kutta optimizer (RUN), Colony Predation Algorithm (CPA), and Harris Hawks Optimization (HHO)). Their achievements in various continuous and discrete problems have been presented in the literature. In this study, a new natural inspiration-based meta-heuristic optimization algorithm called Aquila Optimizer (AO) is introduced. The AO algorithm was first

Table 19

The comparisons of BAO-T and BAO-CM with other algorithms (according to standard deviation results).

Id	BEO	WOS	PSOS	FPS	CSAS	BAS	CGMA	GHSOS	BAO-T	BAO-CM
P21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P25	0.00	0.00	0.00	0.00	0.00	0.00	–	–	0.00	0.00
P26	0.00	5062.05	0.00	0.00	0.00	0.00	–	0.00	0.00	0.00
P27	0.00	9133.17	0.00	0.00	12530.6	0.00	–	0.00	0.00	0.00
P28	0.00	0.00	0.00	0.00	0.00	0.00	–	–	0.00	0.00
P29	0.00	0.00	0.00	0.00	23341.7	–	–	–	0.00	0.00
P30	0.00	0.00	0.00	0.00	0.00	–	–	–	22847.8	0.00
P31	0.00	4132.236	0.00	0.00	0.00	0.00	–	0.00	17974.75	0.00
P32	0.00	42733.237	0.00	0.00	0.00	0.00	–	0.00	44678.91	0.00
P33	0.00	0.00	0.00	0.00	0.00	0.00	–	–	28382.66	0.00
P34	0.00	0.00	0.00	0.00	0.00	4879.200	0.00	0.00	21850.95	0.00
P35	0.00	0.00	16054.133	0.00	0.00	0.00	–	–	9250.46	0.00
P36	0.00	0.00	15820.500	0.00	0.00	0.00	–	0.00	40736.37	0.00
P37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	31356.65	0.00
P38	0.00	7182.320	0.00	0.00	0.00	580.400	0.00	0.00	58919.13	0.00
P39	0.00	4913.303	6305.624	0.00	0.00	0.00	0.00	0.00	36907.38	0.00
P40	0.00	8573.682	0.00	0.00	0.00	0.00	–	–	35604.71	0.00
P41	0.00	8573.682	0.00	0.00	0.00	0.00	–	–	43821.19	0.00
P42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	37218.2	0.00
P43	0.00	0.00	0.00	0.00	0.00	0.00	–	0.00	29766.15	0.00
P44	2795.2	0.00	1879.624	1879.624	1879.624	210.6	–	–	45549.72	0.00
P45	0.00	14162.599	0.00	0.00	0.00	0.00	–	–	44737.25	0.00

***Bold** values indicate the optimal results.**Table 20**

The comparisons of BAO-T and BAO-CM with other algorithms (according to gap results).

Id	BEO	WOS	PSOS	FPS	CSAS	BAS	CGMA	GHSOS	BAO-T	BAO-CM
P21	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P22	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P25	0.00	0.00	0.00	0.00	0.00	0.00	–	–	0.00	0.00
P26	0.00	0.01	0.00	0.00	0.00	0.00	0.09	0.00	0.00	0.00
P27	0.00	0.06	0.00	0.00	0.15	0.00	0.05	0.00	0.00	0.00
P28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P29	0.00	0.00	0.00	0.00	0.029	0.00	0.00	–	0.00	0.00
P30	0.00	0.00	0.00	0.00	0.00	–	–	0.00	0.31	0.00
P31	0.00	0.22	0.00	0.00	0.00	0.00	0.25	0.00	0.36	0.00
P32	0.00	0.20	0.00	0.00	0.00	0.00	0.19	0.00	0.57	0.00
P33	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.34	0.00
P34	0.00	0.12	0.00	0.00	0.00	0.019	0.039	0.00	0.24	0.00
P35	0.00	0.00	0.22	0.00	0.00	0.00	0.015	0.017	0.13	0.00
P36	0.00	0.00	0.05	0.00	0.00	0.00	0.00	0.00	0.89	0.00
P37	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.80	0.00
P38	0.00	0.062	0.00	0.00	0.00	0.0027	0.013	0.00	0.85	0.00
P39	0.00	0.13	0.04	0.00	0.00	0.00	0.12	0.00	1.14	0.00
P40	0.00	0.06	0.00	0.00	0.00	0.00	0.0004	0.005	0.72	0.00
P41	0.00	0.27	0.00	0.00	0.00	0.00	0.08	0.019	1.37	0.00
P42	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.33	0.00
P43	0.00	0.00	0.00	0.00	0.00	0.00	0.0012	0.00	1.35	0.00
P44	0.00361	0.044	0.037	0.044	0.044	0.035	0.05	0.00089	1.27	0.0017
P45	0.00	0.07	0.00	0.00	0.00	0.00	–	–	1.29	0.00
Final rank	1	5	3	1	2	3	6	4	7	1

***Bold** values indicate the optimal results.

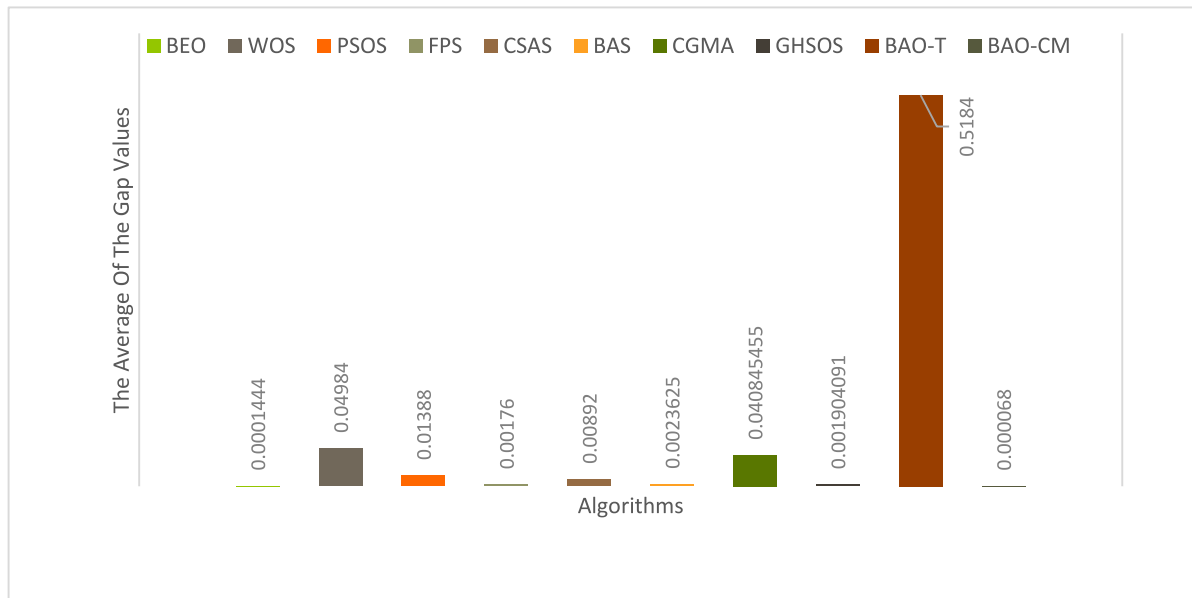
proposed by Abualigah et al. in 2021. The AO algorithm simulates the behavior of Aquila during the hunt, showing the actions of each step of the hunt. Due to its hunting bravery, Aquila is one of the most studied birds globally. Male Aquila got more prey when solo-hunting. Aquila utilizes their speed and sharp talons to hunt squirrels, rabbits, and many other animals. They have even been recognized as an attacker of full-grown deer. The next most notable animal in the diet of Aquila is the ground squirrel. The Aquila Optimizer (AO) algorithm is represented by four methods. These are 1- select the search area high and fly with a vertical stoop; 2- exploring within a diverge search space by contour flight with a short glide attack; 3- exploiting within a converge search space by low flight with a slow descent attack; 4- swooping by walk and grab prey. In this paper, the AO algorithm has been updated again to

solve binary optimization problems. A new Binary AO (BAO) algorithm has been proposed in the literature. The knapsack problem, which is often chosen as a binary optimization problem, has been solved with BAO and the success of BAO has been demonstrated. Three different knapsack datasets were solved in this study. These knapsack problems consist of low, medium, and large-scale dimensions datasets. Thus, the success of BAO has been proven not only in low dimensions but also in large-scale knapsack problems. First of all, the parameter settings affecting the success of BAO were examined in detail and the most appropriate parameter values were selected. In addition, in the first stage of the BAO experimental section, eight different transfer functions were tested and the most successful transfer function was determined (BAO-T). Transfer functions play an important role in the conversion

Table 21

The comparisons of BAO-T and BAO-CM with other algorithms (according to one to one).

Problems		Algorithms									
Id	Name	BEO	WOS	PSOS	FPS	CSAS	BAS	CGMA	GHSOS	BAO-T	BAO-CM
P21	Kp_8a	=	=	=	=	=	=	=	=	=	=
P22	Kp_8b	=	=	=	=	=	=	=	=	=	=
P23	Kp_8c	=	=	=	=	=	=	=	=	=	=
P24	Kp_8d	=	=	=	=	=	=	=	=	=	=
P25	Kp_8e	=	=	=	=	=	=	=	=	=	=
P26	Kp_12a	=	=	=	=	=	=	=	=	=	=
P27	Kp_12b	=	=	=	=	=	=	=	=	=	=
P28	Kp_12c	=	=	=	=	=	=	=	=	=	=
P29	Kp_12d	=	=	=	=	=	=	=	=	=	=
P30	Kp_12e	=	=	=	=	=	=	=	=	=	=
P31	Kp_16a	=	=	=	=	=	=	=	=	=	=
P32	Kp_16b	=	=	=	=	=	=	=	=	=	=
P33	Kp_16c	=	=	=	=	=	=	=	=	=	=
P34	Kp_16d	=	=	=	=	=	=	=	=	=	=
P35	Kp_16e	=	=	=	=	=	=	=	=	=	=
P36	Kp_20a	=	=	=	=	=	=	=	=	=	=
P37	Kp_20b	=	=	=	=	=	=	=	=	=	=
P38	Kp_20c	=	=	=	=	=	=	=	=	=	=
P39	Kp_20d	=	=	=	=	=	=	=	=	=	=
P40	Kp_20e	=	=	=	=	=	=	=	=	=	=
P41	Kp_24a	=	=	=	=	=	=	=	=	=	=
P42	Kp_24b	=	=	=	=	=	=	=	=	=	=
P43	Kp_24c	=	=	=	=	=	=	=	=	=	=
P44	Kp_24d	=	=	=	=	=	=	=	=	=	=
P45	Kp_24e	=	=	=	=	=	=	=	=	=	=
W/D/L		0/24/1	0/14/11	0/21/4	0/24/1	0/14/11	0/20/5	0/10/15	1/18/6	0/9/16	0/24/1

**Fig. 14.** The comparisons of the gap values of the datasets from P21 to P45.

of continuous search space to binary space. Eight popular S and V-shaped transfer functions, mostly selected in the literature, are used for binary conversion in this study. Then, BAO-T was developed by using crossover and mutation techniques in the candidate solution production stage of BAO-T and a BAO-CM variation was created. BAO-T and BAO-CM variations were compared according to mean, best, worst, standard deviation, and gap comparison criteria, and the success of newly added techniques on performance was demonstrated. Comparisons with graphics are explained in detail. The superior success of BAO-CM has been proven by comparing BAO-T and BAO-CM variations with recent algorithms selected from the literature. The results showed that the BAO-CM algorithm is a successful alternative algorithm for solving binary optimization problems.

In future studies, it is thought that the success of BAO will be tested in different binary optimization problems. In addition, transfer

functions (such as U and Taper-shaped transfer functions) used in the literature other than S and V-shaped transfer functions will be examined and their success on BAO will be examined.

CRedit authorship contribution statement

Emine Baş: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 22

The comparisons of BAO-T and BAO-CM with other algorithms (according to mean results).

Id	BPSO	MBPSO	NGHS	DGHS	S-bAFSA	CGMA	BinEHO	BAO-T	BAO-CM
P21	3921857.19	3924400	3924400	3924400	3924400	3924400	3924400	3924400	3924400
P22	3807911.86	3813669	3813669	3813669	3813669	3813669	3813669	3813669	3813669
P23	3328608.71	3347452	3347452	3347452	3347452	3347452	3347452	3347452	3347452
P24	4186088.27	4187707	4187707	4187707	4187707	4187707	4187707	4187707	4187707
P25	4932737.28	4954571.72	4955555	4955555	4955555	4955555	4955555	4955555	4955555
P26	5683694.29	5688552.41	5687724.44	5687460.74	5688498.02	5688887	5688757.34	5688887	5688887
P27	6478582.96	6493130.57	6486450.92	6498597	6498597	6498597	6498597	6498597	6498597
P28	5166957.08	5170493.30	5165079.86	5165079.80	5170626	5170626	5170626	5170626	5170626
P29	6989842.73	6992144.26	6991971.10	6992404	6992404	6992404	6992404	6992404	6992404
P30	5316879.59	5337472	5273974.06	5335555.92	5337472	5337472	5337472	5320706	5337472
P31	7834900.26	7843073.29	7795177.56	7850983	7850983	7850983	7850224.60	7822759	7850983
P32	9334408.62	9350353.39	9253153.36	9352998	9352998	9352998	9352998	9299500	9352998
P33	9118837.47	9144118.38	9055853.46	9151147	9151147	9151147	9151147	9119861	9151147
P34	9321705.87	9321705.87	9310913.74	9316797.66	9321195.66	9345229.60	9348889	9326542	9348889
P35	7758572.21	7764131.81	7738799.78	9316797.66	7761387.72	7769117	7766509.36	7759291	7769117
P36	10707360.91	10720314.03	10592304.82	10727049	10727049	10727049	10727049	10631817	10727049
P37	9791306.65	9805480.48	9728979.66	9818261	9818261	9818261	9818261	9740051	9818261
P38	10703423.34	10710947.05	10585114.14	10709663.64	10710831.94	10712553.74	10713587.70	10623099	10714023
P39	8910152.57	8923712.21	8859297.78	8916496.80	8917872.80	8918423.20	8929156	8827026	8929156
P40	9349546.98	9355930.35	9324698.84	9357518.34	9357331.86	9357969	9357751.44	9290988	9357969
P41	13510432.96	13532060.07	13508995.14	13527887.56	13524874.30	13538097.18	13546986.14	13363882	13549094
P42	12205346.16	12223442.61	12160953.58	12233713	12233713	12233713	12233713	12071440	12233713
P43	12427880.56	12443349.03	12424467.44	12448158.86	12448158.86	12448618.30	12448780	12280388	12448780
P44	11792064.76	11803712.38	11736314.98	11810261.56	11810261.56	11808614.90	11810682.68	11665735	11815109.72
P45	13922797.55	13932526.16	13827901.40	13940099	13940099	13940099	13940099	13759686	13940099
Final rank	8	7	7	5	4	3	2	6	1

***Bold** values indicate the optimal results.**Table 23**

The comparisons of BAO-T and BAO-CM with other algorithms (according to gap results).

Id	BPSO	MBPSO	NGHS	DGHS	S-bAFSA	CGMA	BinEHO	BAO-T	BAO-CM
P21	0.48E-02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P22	1.51E-01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P23	5.63E-01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P24	3.87E-02	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P25	4.60E-01	1.98E-02	0.00	0.00	0.00	0.00	0.00	0.00	0.00
P26	9.13E-02	5.88E-03	2.04E-02	2.51E-02	6.84E-03	0.00	2.28E-03	0.00	0.00
P27	3.08E-01	8.41E-02	1.87E-01	0.00	0.00	0.00	0.00	0.00	0.00
P28	7.10E-02	2.57E-03	1.07E-01	1.07E-01	0.00	0.00	0.00	0.00	0.00
P29	3.66E-02	3.71E-03	6.19E-03	0.00	0.00	0.00	0.00	0.00	0.00
P30	3.86E-01	0.00	1.19E+00	3.59E-02	0.00	0.00	0.00	0.31	0.00
P31	2.05E-01	1.01E-01	7.11E-01	0.00	0.00	0.00	9.66E-03	0.36	0.00
P32	1.99E-01	2.83E-02	1.07E+00	0.00	0.00	0.00	0.00	0.57	0.00
P33	3.53E-01	7.68E-02	1.04E+00	0.00	0.00	0.00	0.00	0.34	0.00
P34	2.91E-01	1.17E-01	4.06E-01	3.43E-01	2.96E-01	3.91E-02	0.00	0.24	0.00
P35	1.36E-01	6.42E-02	3.90E-01	2.68E-01	9.95E-02	1.53E-02	3.36E-02	0.13	0.00
P36	1.84E-01	6.28E-02	1.26E+00	0.00	0.00	0.00	0.00	0.89	0.00
P37	2.75E-01	1.30E-01	9.09E-01	0.00	0.00	0.00	0.00	0.80	0.00
P38	9.89E-02	2.87E-02	1.20E+00	4.07E-02	2.98E-02	1.37E-02	4.06E-03	0.85	0.00
P39	2.13E-01	6.10E-02	7.82E-01	1.42E-01	1.26E-01	1.20E-01	0.00	1.14	0.00
P40	9.00E-02	2.18E-02	3.56E-01	4.82E-03	6.81E-03	4.98E-04	2.32E-03	0.72	0.00
P41	2.85E-01	1.26E-01	2.96E-01	1.57E-01	1.79E-01	8.12E-02	1.56E-02	1.37	0.00
P42	2.32E-01	8.40E-02	5.95E-01	0.00	0.00	0.00	0.00	1.33	0.00
P43	1.68E-01	4.36E-02	1.95E-01	4.99E-03	1.85E-03	1.30E-03	0.00	1.35	0.00
P44	1.97E-01	9.82E-02	6.69E-01	4.28E-02	3.74E-02	5.67E-02	3.92E-02	1.27	0.0017
P45	1.24E-01	5.43E-02	8.05E-01	0.00	0.00	0.00	0.00	1.29	0.00
Final rank	8	7	7	5	4	3	2	6	1

***Bold** values indicate the optimal results.**Table 24**

The comparisons of BAO-T and BAO-CM with other algorithms (according to one to one).

Problems		Algorithms								
Id	Name	BPSO	MBPSO	NGHS	DGHS	S-bAFSA	CGMA	BinEHO	BAO-T	BAO-CM
P21	Kp_8a	-	=	=	=	=	=	=	=	=
P22	Kp_8b	-	=	=	=	=	=	=	=	=
P23	Kp_8c	-	=	=	=	=	=	=	=	=
P24	Kp_8d	-	=	=	=	=	=	=	=	=
P25	Kp_8e	-	-	=	=	=	=	=	=	=
P26	Kp_12a	-	-	-	-	-	=	-	=	=

(continued on next page)

Table 24 (continued).

Problems		Algorithms								
Id	Name	BPSO	MBPSO	NGHS	DGHS	S-bAFSA	CGMA	BinEHO	BAO-T	BAO-CM
P27	Kp_12b	-	-	-	=	=	=	=	=	=
P28	Kp_12c	-	-	-	-	=	=	=	=	=
P29	Kp_12d	-	-	-	=	=	=	=	=	=
P30	Kp_12e	-	=	-	=	=	=	=	-	=
P31	Kp_16a	-	-	-	=	=	=	-	-	=
P32	Kp_16b	-	-	-	=	=	=	=	-	=
P33	Kp_16c	-	-	-	=	=	=	=	-	=
P34	Kp_16d	-	-	-	-	-	-	=	-	=
P35	Kp_16e	-	-	-	-	-	-	-	-	=
P36	Kp_20a	-	-	-	=	=	=	=	-	=
P37	Kp_20b	-	-	-	=	=	=	=	-	=
P38	Kp_20c	-	-	-	-	-	-	-	-	+
P39	Kp_20d	-	-	-	-	-	-	=	-	=
P40	Kp_20e	-	-	-	-	-	-	-	-	+
P41	Kp_24a	-	-	-	-	-	-	-	-	+
P42	Kp_24b	-	-	-	=	=	=	=	-	=
P43	Kp_24c	-	-	-	-	-	-	=	-	=
P44	Kp_24d	-	-	-	-	-	-	-	-	+
P45	Kp_24e	-	-	-	=	=	=	=	-	=
W/D/L		0/0/25	0/5/20	0/5/20	0/14/11	0/16/9	0/17/8	0/18/7	0/9/16	4/21/0

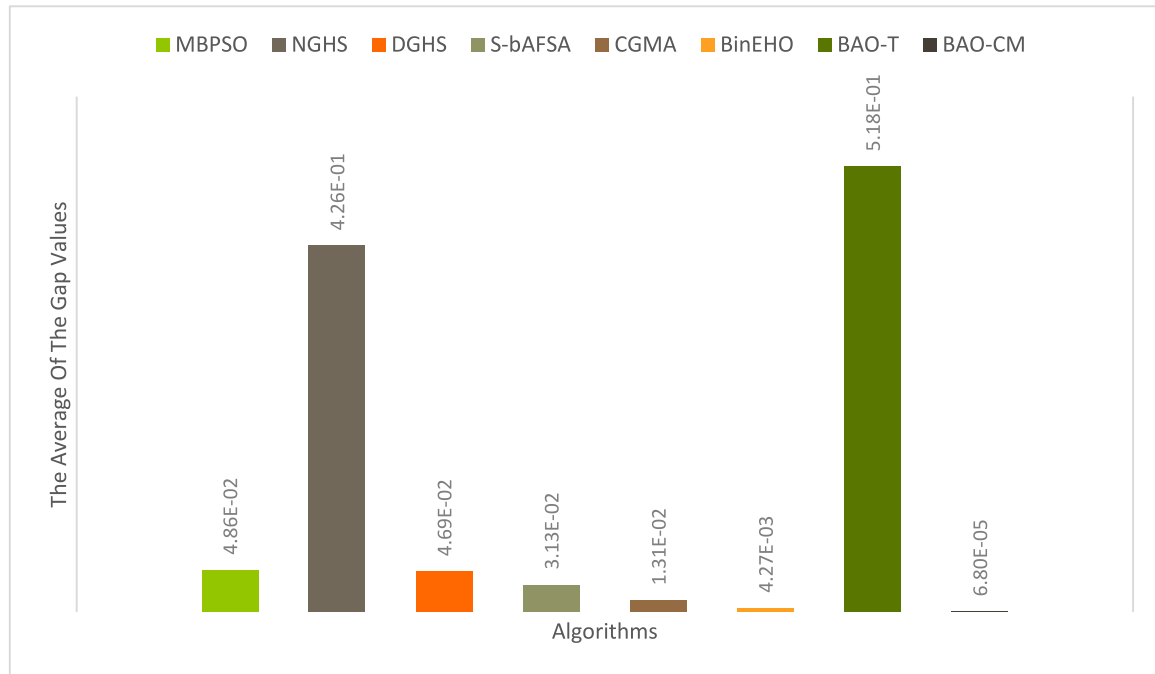


Fig. 15. The comparisons of the gap values of the datasets from P21 to P45.

Table 25

The comparisons of BAO-CM with other algorithms on the third group knapsack datasets (according to mean results).

Id	Name	Optimum	EOS2	SA	GA	IGA-SA	GSA	BB	BAO-CM
P46	Kp1_100	9147	9147	9147	9147	8575	2983	8026	9147
P47	Kp1_200	11238	11238	10163	102.340	8576	9865	10438	11238
P48	Kp1_500	28857	28857	21390	102340	12072	9865	28043	28857
P49	Kp1_1000	54503	54503	36719	130	14563	14927	53397	54503
P50	Kp1_2000	110625	110625	95739	102340	27645	25579	109679	110625
P51	Kp1_5000	276457	274358	150731	102340	49306	39677	275720	276457
P52	Kp2_100	1514	1514	1486	1158	1217	1041	1440	1514
P53	Kp2_200	1634	1634	1537	1306	1347	1073	1603	1634
P54	Kp2_500	4566	4564.4	3744	3701	3038	2951	4484	4566
P55	Kp2_1000	9052	9050.8	6831	7912	5435	5675	9006	9042.75
P56	Kp2_2000	18051	17497	12780	15887	10938	11064	17794	18027.45
P57	Kp2_5000	44356	44298	29220	37746	27387	25448	44198	44332
P58	Kp3_100	2397	2397	2296	2091	1481	1095	2268	2397

(continued on next page)

Table 25 (continued).

Id	Name	Optimum	EOS2	SA	GA	IGA-SA	GSA	BB	BAO-CM
P59	Kp3_200	2697	2697	2594	25.319	1495	1095	2542	2697
P60	Kp3_500	7117	7117	6103	6517	3412	2916	6995	7117
P61	Kp3_1000	14390	14390	11789	13090	5589	6290	14271	14390
P62	Kp3_2000	28919	28919	22482	25319	10818	12312	28726	28919
P63	Kp3_5000	72505	71984	53672	61904	27304	30302	72.345	72498.95
Final rank			2	3	4	5	5	5	1

***Bold** values indicate the optimal results.

Table 26

The comparisons of BAO-CM with other algorithms on the third group knapsack datasets (according to gap results).

Id	Name	Optimum	EOS2	SA	GA	IGA-SA	GSA	BB	BAO-CM
P46	Kp1_100	9147	0.00	0.00	0.00	6.253	67.388	12.255	0.00
P47	Kp1_200	11238	0.00	9.566	-89.019	23.687	12.217	7.119	0.00
P48	Kp1_500	28.857	0.00	25.876	-71.803	58.166	65.814	2.821	0.00
P49	Kp1_1000	54503	0.00	32.629	99.761	73.280	72.613	2.029	0.00
P50	Kp1_2000	110625	0.00	13.419	8.050	75.000	76.868	0.813	0.00
P51	Kp1_5000	276457	0.759	45.478	62.982	82.165	85.648	0.267	0.00
P52	Kp2_100	1514	0.00	1.849	23.514	19.617	31.242	4.888	0.00
P53	Kp2_200	1634	0.00	5.936	25.115	17.564	34.333	1.897	0.00
P54	Kp2_500	4566	0.035	18.03	23.372	33.465	35.370	1.796	0.00
P55	Kp2_1000	9052	0.013	24.536	12.594	39.958	37.307	0.508	0.102
P56	Kp2_2000	18051	3.069	29.201	13.621	39.405	38.707	1.424	0.131
P57	Kp2_5000	44356	0.131	34.124	14.902	38.256	42.628	0.356	0.054
P58	Kp3_100	2397	0.00	4.214	12.766	38.214	54.318	5.382	0.00
P59	Kp3_200	2697	0.00	3.819	-838.784	44.568	59.399	5.747	0.00
P60	Kp3_500	7117	0.00	14.248	8.431	52.058	59.028	1.714	0.00
P61	Kp3_1000	14390	0.00	18.075	9.034	61.161	56.289	0.827	0.00
P62	Kp3_2000	28919	0.00	22.259	12.449	62.592	57.426	0.667	0.00
P63	Kp3_5000	72505	0.719	25.975	14.621	62.342	58.207	0.221	0.008
Final rank			2	3	4	5	5	5	1

***Bold** values indicate the optimal results.

**Fig. 16.** The comparisons of the gap values of the datasets from P46 to P63.

Table 27

The comparisons of BAO-CM with other algorithms on the third group knapsack datasets (according to one to one).

Problems		Algorithms						
Id	Name	EOS2	SA	GA	IGA-SA	GSA	BB	BAO-CM
P46	Kp1_100	=	=	=	-	-	-	=
P47	Kp1_200	=	-	-	-	-	-	=
P48	Kp1_500	=	-	-	-	-	-	=
P49	Kp1_1000	=	-	-	-	-	-	=
P50	Kp1_2000	=	-	-	-	-	-	=
P51	Kp1_5000	-	-	-	-	-	-	+
P52	Kp2_100	=	-	-	-	-	-	=
P53	Kp2_200	=	-	-	-	-	-	=
P54	Kp2_500	-	-	-	-	-	-	+
P55	Kp2_1000	+	-	-	-	-	-	-
P56	Kp2_2000	-	-	-	-	-	-	+
P57	Kp2_5000	-	-	-	-	-	-	+
P58	Kp3_100	=	-	-	-	-	-	=
P59	Kp3_200	=	-	-	-	-	-	=
P60	Kp3_500	=	-	-	-	-	-	=
P61	Kp3_1000	=	-	-	-	-	-	=
P62	Kp3_2000	=	-	-	-	-	-	=
P63	Kp3_5000	-	-	-	-	-	-	+
W/D/L		1/12/5	0/1/17	0/1/17	0/0/18	0/0/18	0/0/18	5/12/1

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